

IC Boards Question Bank

138 single-best-answer questions with explanations

Part I: questions grouped by blueprint domain, answers withheld.

Part II: answer key on one page.

Part III: answers with explanations, in the same order.

Question numbers (Q1–Q138) match the morning emails and the online quiz.

ABIM Interventional Cardiology exam · October 6, 2026 · Built October 3, 2026

Part I. Questions

Case selection 22 · Procedural technique 15 · Pharmacology 18 · Imaging 8 · Physiology and basic science 9 · Complications 11 · Hemodynamics and mechanical support 11 · Structural heart 15 · Peripheral and noncardiac vascular 7 · Anatomy 8 · Radiation safety 4 · Statistics, quality and regulation 10

Case selection

Q49. A 66-year-old with STEMI and cardiogenic shock has a 100% proximal LAD (culprit), 80% mid RCA and 70% OM1. What is the recommended revascularization strategy at the index procedure?

- A. PCI of all three lesions immediately
- B. Culprit-only PCI with staged evaluation of the non-culprit lesions
- C. Emergency CABG
- D. Fibrinolysis followed by CABG
- E. PCI of the LAD and RCA, deferring the OM

Q50. A 60-year-old diabetic with stable angina has three-vessel disease (SYNTAX 29), EF 45%, and is a good surgical candidate. What does the evidence favor?

- A. Multivessel PCI with DES
- B. CABG
- C. Medical therapy only
- D. Hybrid revascularization
- E. FFR-guided PCI of only the ischemic lesions

Q51. A 72-year-old presents 30 hours after onset of chest pain that has now resolved, with Q waves and persistent ST elevation in inferior leads, TIMI 0 flow in the RCA, hemodynamically stable, no arrhythmias or heart failure. What is the recommended management?

- A. Primary PCI immediately
- B. Fibrinolysis
- C. Medical therapy; routine PCI of a totally occluded infarct artery >24 h in stable asymptomatic patients has no benefit
- D. Elective PCI in 1 week
- E. CABG

Q52. A 55-year-old man with NSTEMI is hemodynamically stable, pain-free, with dynamic ST changes, troponin rise and GRACE score 155. When should angiography occur?

- A. Immediately (<2 h)
- B. Within 24 h
- C. Within 72 h
- D. Ischemia-guided only if non-invasive testing is positive
- E. After 5 days of anticoagulation

Q53. A 38-year-old postpartum woman presents with NSTEMI; angiography shows a long smooth narrowing of the mid-distal LAD with TIMI 3 flow and no atherosclerosis elsewhere. What is the best management?

- A. PCI with a long DES
- B. Intracoronary thrombolysis
- C. Conservative management with beta-blocker and aspirin, and screen for fibromuscular dysplasia
- D. CABG
- E. Rotational atherectomy

Q54. A 70-year-old with stable angina on two antianginals has moderate ischemia on nuclear imaging; angiography shows 70% proximal LAD and 80% RCA, normal EF, no LM. Based on ISCHEMIA, what should the patient be told?

- A. Revascularization reduces the risk of death compared with medical therapy
- B. Revascularization does not reduce death or MI overall but improves angina and quality of life
- C. Medical therapy is unsafe with this ischemic burden
- D. CABG is mandatory
- E. PCI of the LAD prolongs survival

Q55. Left main 70% stenosis (isolated, SYNTAX 18) in a 68-year-old with STS predicted mortality 1%. Which statement is best supported?

- A. PCI is contraindicated because CABG is superior in all LM disease
- B. PCI with DES is a reasonable alternative to CABG (class 2a) with equivalent mortality in low-intermediate SYNTAX
- C. PCI is preferred over CABG (class 1)
- D. Medical therapy alone is appropriate
- E. CABG is contraindicated at this low risk

Q79. A 62-year-old with HBR (eGFR 25, prior GI bleed on anticoagulation, on apixaban for AF) requires PCI with DES for stable angina. Which DAPT strategy is best supported?

- A. Aspirin + clopidogrel + apixaban for 12 months
- B. Aspirin + clopidogrel + apixaban for ≤ 1 week, then clopidogrel + apixaban for 6–12 months, then apixaban alone
- C. Prasugrel + apixaban for 12 months
- D. Aspirin + apixaban only
- E. Stop apixaban and use DAPT for 12 months

Q82. A 70-year-old with LVEF 30% and complex three-vessel disease is scheduled for PCI. Based on CHIP-BCIS3 (2026), what is the effect of routine prophylactic Impella CP support?

- A. Reduced MACE at 1 year
- B. No benefit, with higher cardiovascular death in the Impella arm
- C. Reduced periprocedural MI only
- D. Reduced need for emergency CABG
- E. Improved LVEF recovery at 6 months

Q94. In STEMI with multivessel disease and no shock, what did MULTISTARS AMI show about the timing of non-culprit PCI?

- A. Staged PCI at 19–45 days was superior to immediate complete PCI
- B. Immediate multivessel PCI was non-inferior to staged PCI, with fewer MIs and unplanned revascularizations
- C. Immediate multivessel PCI increased mortality
- D. Non-culprit PCI should be deferred to 6 months
- E. FFR must guide all non-culprit PCI

Q103. Day 4 after an inferior STEMI treated late, a patient develops sudden pulmonary edema and hypotension. A soft systolic murmur is heard. PA catheter shows giant v waves on the wedge tracing and no oxygen step-up between RA and PA. What is the diagnosis and the immediate plan?

- A. Ventricular septal rupture; percutaneous closure
- B. Posteromedial papillary muscle rupture; afterload reduction, IABP/Impella bridge and emergent surgery
- C. Free-wall rupture; pericardiocentesis
- D. Acute stent thrombosis; repeat PCI
- E. RV infarction; volume loading

Q104. A 70-year-old woman presents 3 days after an untreated anterior STEMI with a new harsh holosystolic murmur and shock. Right heart catheterization shows RA saturation 62% and PA saturation 80%. Which is correct?

- A. Papillary muscle rupture; giant v waves expected
- B. Ventricular septal rupture at the apex; highest shock mortality; IABP/afterload reduction and surgery, with device closure for residual or as a bridge
- C. Free-wall rupture; tamponade physiology
- D. Severe MR from papillary dysfunction; medical therapy
- E. Dynamic LVOT obstruction; fluids and beta-blocker

Q105. A patient with an acute inferior STEMI has hypotension, clear lungs, elevated JVP and ST elevation in V4R. After primary PCI of the RCA the blood pressure remains 80/50 with RA 18 and PCWP 10. What is the next best step?

- A. Intravenous nitroglycerin
- B. Furosemide
- C. Cautious volume loading, maintain AV synchrony, dobutamine if no response, and consider RV mechanical support
- D. Intra-aortic balloon pump as first-line
- E. Beta-blocker to reduce myocardial oxygen demand

Q106. A 52-year-old smoker with typical angina, no known CAD, and an interpretable ECG who can exercise is being evaluated. Per the 2021 chest pain and 2023 CCD guidelines, which initial test is most appropriate for an intermediate-risk patient under 65 not on optimal medical therapy?

- A. Invasive coronary angiography
- B. Coronary CT angiography (with CT-FFR for 40–90% lesions if needed)
- C. Routine stress SPECT regardless of age
- D. Coronary artery calcium score alone
- E. No testing

Q107. A patient with STEMI presents to a hospital without PCI capability. The nearest PCI center is 2 hours away by ambulance. Symptom onset was 90 minutes ago. What is the correct reperfusion strategy?

- A. Transfer for primary PCI regardless of delay
- B. Fibrinolysis within 30 minutes of arrival, then transfer to a PCI center with angiography within 2–24 hours
- C. Fibrinolysis and immediate PCI on arrival (facilitated PCI)
- D. Fibrinolysis and repeat lysis if ST does not resolve
- E. Aspirin, heparin and transfer for angiography in 48 hours

Q108. A 61-year-old with chest pain has a normal ECG and two negative high-sensitivity troponins at 0 and 2 hours; HEART score is low. What is the appropriate strategy?

- A. Immediate invasive angiography
- B. Early invasive strategy within 24 hours
- C. Selective (ischemia-guided) approach; early invasive evaluation is Class 3 for low likelihood of ACS
- D. Empiric ticagrelor load and admission
- E. Fibrinolysis

Q110. A 68-year-old had an everolimus-eluting stent placed 4 months ago for stable angina and now needs a non-urgent but time-sensitive resection of a colon cancer. He is on aspirin and clopidogrel. What is the recommended perioperative plan?

- A. Proceed now, stopping both antiplatelets 7 days before surgery
- B. Delay all surgery until 12 months after stenting
- C. Proceed after 3 months is acceptable for time-sensitive surgery; continue aspirin, hold clopidogrel 5 days, resume within 24–48 h
- D. Bridge with therapeutic enoxaparin
- E. Perform preoperative stress testing and revascularize any ischemia first

Q115. Sixteen hours after an anterior STEMI treated with primary PCI, a patient develops a new right bundle branch block with left anterior fascicular block and intermittent Mobitz II second-degree AV block, heart rate 44, blood pressure 92/60. What is the appropriate management?

- A. Observation; this block is at the AV node and will resolve
- B. Atropine and observation
- C. Prophylactic temporary transvenous pacing because the block is infranodal with an unreliable escape
- D. Isoproterenol infusion
- E. Immediate permanent pacemaker

Q122. A 67-year-old with mild stable angina controlled on two drugs has moderate-to-severe ischemia (12% of the myocardium) on PET; CT angiography excludes left main disease and shows three-vessel disease with preserved EF. Based on ISCHEMIA, which statement is correct?

- A. Routine invasive management reduces death and MI compared with optimal medical therapy
- B. An initial invasive strategy does not reduce death or MI over about 3.5 years but improves angina control; medical therapy with revascularization for refractory symptoms is appropriate
- C. Invasive management is contraindicated
- D. CABG is mandatory for three-vessel disease
- E. The stress imaging result alone mandates angiography

Q123. A 70-year-old with ischemic cardiomyopathy (LVEF 28%), three-vessel disease with viable myocardium on CMR and no angina is on optimal medical therapy with an ICD. Based on REVIVED-BCIS2, what does PCI offer?

- A. Reduced all-cause death or heart-failure hospitalization
- B. Improved LVEF at 12 months
- C. No reduction in death or heart-failure hospitalization and no sustained improvement in LVEF or quality of life compared with medical therapy alone
- D. Equivalence to CABG as shown in STICH
- E. Fewer ventricular arrhythmias

Q124. A 58-year-old presents with an anterior STEMI to a hospital 3 hours from a PCI center. Which finding is an ABSOLUTE contraindication to fibrinolysis?

- A. Blood pressure 185/105 mmHg on arrival
- B. Ischemic stroke 8 months ago
- C. Ischemic stroke 6 weeks ago
- D. Current warfarin use with INR 2.5
- E. Pregnancy

Q125. A 63-year-old with insulin-treated diabetes, LVEF 50%, and three-vessel disease (SYNTAX 24) without left main involvement has class II angina. Which revascularization is recommended?

- A. Multivessel PCI with DES because the SYNTAX score is ≤ 32
- B. CABG with a LIMA to the LAD (class 1 over PCI in diabetic multivessel disease)
- C. Hybrid revascularization
- D. Medical therapy only; revascularization is not indicated
- E. PCI of the LAD only

Procedural technique

Q56. After provisional stenting of the LAD across a Medina 1,1,1 bifurcation with a 2.5 mm diagonal, the diagonal shows 80% ostial stenosis with TIMI 3 flow and no dissection; the patient is asymptomatic. Which is the most appropriate next step?

- A. Immediate TAP stenting of the diagonal
- B. Proximal optimization technique, re-wire the diagonal through a distal cell and perform kissing balloon inflation
- C. Leave as is; ostial side branch pinching is cosmetic
- D. Crush stent
- E. Retrograde wiring of the diagonal via septals

Q57. For a true distal left main bifurcation (Medina 1,1,1) with a 3.5 mm LCx having 15 mm of ostial disease, which strategy has the best randomized evidence?

- A. Provisional single stent
- B. DK crush two-stent technique
- C. Simultaneous kissing stents
- D. T-stenting
- E. Culotte with a 6F guide

Q58. A right coronary CTO has a blunt proximal cap, 25 mm length, moderate calcification, no bend $>45^\circ$, and this is the first attempt. What is the J-CTO score and predicted difficulty?

- A. 1, intermediate
- B. 2, difficult
- C. 3, very difficult
- D. 4, very difficult
- E. 0, easy

Q59. During a retrograde CTO attempt, which collateral pathway carries the highest risk of life-threatening tamponade if perforated, and what is the rule regarding balloon dilation?

- A. Septal collaterals; never dilate
- B. Epicardial collaterals; never dilate
- C. Septal collaterals; dilate with 1.5 mm balloon routinely
- D. Epicardial collaterals; dilate with 2.0 mm balloon
- E. Both can be dilated up to 2.0 mm

Q60. Which is correct regarding rotational atherectomy technique?

- A. Burr-to-artery ratio 0.8 gives the best lumen
- B. Rotational speed 200,000 rpm minimizes heat
- C. Avoid decelerations >5000 rpm and use short pecking runs of 15–20 s
- D. Rotablation is preferred for thrombotic lesions
- E. Rotablation can be done over a standard 0.014 inch workhorse wire

Q61. Which statement about intravascular lithotripsy is correct?

- A. It requires blood clearance with contrast
- B. It is contraindicated in patients with permanent pacemakers
- C. Sonic pressure waves fracture calcium with a 1:1 sized balloon inflated at 4 atm, ~10 pulses per cycle
- D. It is ineffective for deep or eccentric calcium
- E. It causes more slow-flow than rotablation

Q62. A 70% lesion in the body of a 6-year-old degenerated saphenous vein graft to the OM is treated. Which measure most reduces periprocedural MI?

- A. Predilation with an oversized balloon
- B. Distal embolic protection device
- C. Covered stent
- D. Bare metal stent instead of DES
- E. Intracoronary nitroglycerin before stenting

Q63. Two years after DES placement a patient has diffuse in-stent restenosis (Mehran type II). IVUS shows the stent is well expanded with homogeneous neointima. Which treatment is supported by AGENT IDE?

- A. Plain balloon angioplasty
- B. Paclitaxel drug-coated balloon
- C. Bare metal stent
- D. Rotational atherectomy
- E. Brachytherapy first-line

Q64. You plan a two-stent DK crush of a non-LM bifurcation through a 6F guide. Which statement is correct?

- A. 6F cannot accommodate any two-stent technique
- B. Kissing two stents simultaneously requires 8F
- C. Sequential techniques (DK crush, culotte) are feasible through 6F (ID ~0.070 inch); simultaneous kissing stents require 7F or larger
- D. 7F is required for all two-stent techniques
- E. Rotablation with a 2.0 mm burr is possible via 6F

Q80. Which wire is most appropriate as the initial choice for antegrade wire escalation through a CTO with a tapered stump and microchannels?

- A. Confianza Pro 12
- B. Polymer-jacketed tapered wire (e.g., Fielder XT, Fighter)
- C. Standard workhorse wire (e.g., BMW)
- D. Gaia Third
- E. Rotawire

Q89. A patient with refractory angina despite optimal medical therapy has a J-CTO 2 RCA occlusion and no other lesions. What is the ACC/AHA 2021 class of recommendation for CTO PCI, and what did EXPLORE and REVASC show?

- A. Class 1; both showed improved LVEF
- B. Class 2a; both showed reduced mortality
- C. Class 2b; neither showed LVEF improvement
- D. Class 3; harm
- E. Class 2a; improved LVEF in EXPLORE only

Q117. Which statement about endomyocardial biopsy is correct?

- A. Samples are taken from the RV free wall
- B. A pre-existing LBBB is a reason to have temporary pacing available because biopsy can cause RBBB and complete heart block
- C. It is contraindicated in suspected giant cell myocarditis
- D. Perforation occurs in about 10% of cases
- E. Only one sample is needed

Q133. A heavily calcified, tortuous mid-RCA lesion will not accept a stent despite predilation, and the 6F JR4 guide backs out of the ostium with each push. Which strategy best improves deliverability?

- A. Switch to a 5F guide for deeper intubation
- B. Use a guide-extension catheter advanced over an anchor balloon or balloon shaft, add a buddy wire, and consider a more supportive shape (AL1) or 7F guide after further lesion preparation
- C. Push harder with the stent
- D. Abandon the procedure and refer for CABG
- E. Exchange for a hydrophilic wire only

Q134. When stenting an aorto-ostial RCA lesion, which technique minimizes geographic miss and later ostial restenosis?

- A. Position the stent entirely inside the vessel to avoid any aortic protrusion
- B. Disengage the guide slightly, position the stent in a view that opens the ostium (LAO or lateral) so it protrudes 1–2 mm into the aorta, deploy, then flare the proximal struts with a short larger balloon partly in the aorta; the Szabo anchor-wire technique is an alternative
- C. Deploy the stent 5 mm into the aorta and trim it with a cutting balloon
- D. Use a drug-coated balloon only
- E. Always perform rotational atherectomy first

Q135. After provisional stenting of the LAD across a 2.0 mm diagonal, the diagonal ostium shows a 60% pinch with TIMI 3 flow; the patient is asymptomatic without ECG change. What is the recommended next step?

- A. Routine final kissing balloon inflation
- B. Proximal optimization (POT) and leave the diagonal alone; rewire and dilate only for TIMI flow <3, dissection, or ischemia, or for a large side branch
- C. Convert to a culotte
- D. T-stent the diagonal
- E. Rotational atherectomy of the diagonal

Pharmacology

Q12. A 58-year-old presents with NSTEMI, is loaded with aspirin, and coronary angiography is planned within 12 hours. Which P2Y12 strategy is supported by ISAR-REACT 5 and ACCOAST?

- A. Prasugrel 60 mg now in the ED
- B. Ticagrelor 180 mg now, then switch to prasugrel after angiography
- C. Prasugrel 60 mg after coronary anatomy is defined and PCI is chosen
- D. Clopidogrel 600 mg now followed by prasugrel after PCI
- E. Cangrelor infusion started in the ED

Q13. A patient on cangrelor for PCI is to be transitioned to an oral P2Y12 inhibitor. Which is correct?

- A. Give clopidogrel 600 mg at the start of the cangrelor infusion
- B. Give ticagrelor 180 mg during the infusion or immediately after it stops
- C. Give prasugrel 60 mg 2 hours before stopping cangrelor
- D. Any oral agent can be given at any time
- E. Wait 6 hours after stopping cangrelor before any oral agent

Q14. A 55-kg, 78-year-old woman with STEMI undergoes primary PCI. Which P2Y12 regimen is appropriate?

- A. Prasugrel 60 mg then 10 mg daily
- B. Prasugrel 60 mg then 5 mg daily
- C. Ticagrelor 180 mg then 90 mg twice daily
- D. Clopidogrel 300 mg then 75 mg daily
- E. Ticagrelor 180 mg then 60 mg twice daily

Q15. Bivalirudin is chosen for a STEMI PCI. Which practice reduced the excess acute stent thrombosis seen in HORIZONS-AMI?

- A. Adding a 300 mg clopidogrel load
- B. Continuing the full PCI-dose infusion for 2–4 hours after the procedure
- C. Halving the bolus dose
- D. Adding fondaparinux
- E. Using bivalirudin only with femoral access

Q16. A dialysis patient with NSTEMI develops a large thrombus during PCI requiring bail-out GP IIb/IIIa inhibition. Which agent is contraindicated?

- A. Tirofiban at half the infusion rate
- B. Eptifibatide
- C. Abciximab
- D. Cangrelor
- E. All of the above

Q17. A patient with AF (CHA2DS2-VASc 4) undergoes PCI with DES for stable angina. Based on AUGUSTUS and current guidelines, what is the preferred regimen after the first week?

- A. Warfarin + aspirin + clopidogrel for 12 months
- B. Apixaban + clopidogrel, aspirin stopped
- C. Apixaban + ticagrelor + aspirin for 6 months
- D. Apixaban + aspirin, clopidogrel stopped
- E. Rivaroxaban 20 mg + prasugrel

Q18. During PCI with unfractionated heparin (ACT 280 s), you develop a type III perforation. A balloon is inflated across the site and a covered stent is being prepared. Which is the appropriate use of protamine?

- A. Give protamine immediately at 1 mg per 100 U of heparin
- B. Withhold protamine entirely because a stent will be placed
- C. Give protamine after the covered stent has sealed the perforation, if bleeding persists
- D. Give protamine only if the patient has no fish allergy
- E. Give 50 mg protamine IV push

Q19. Which intervention has the strongest evidence for reducing contrast-associated AKI?

- A. N-acetylcysteine 1200 mg twice daily
- B. Sodium bicarbonate infusion
- C. Limiting contrast volume and isotonic saline hydration guided by LVEDP
- D. Prophylactic hemodialysis after the procedure
- E. Iso-osmolar contrast instead of low-osmolar contrast

Q20. A patient with severe no-reflow after SVG PCI has hypotension. Which drug is LEAST likely to help?

- A. Intracoronary nicardipine
- B. Intracoronary adenosine
- C. Intracoronary nitroglycerin
- D. Intracoronary nitroprusside
- E. Intracoronary verapamil

Q21. A patient with prior anaphylactoid contrast reaction needs urgent (within 2 hours) angiography for NSTEMI. What is the best premedication?

- A. Prednisone 50 mg orally now
- B. Hydrocortisone 200 mg IV immediately and every 4 hours until the procedure, plus diphenhydramine 50 mg IV
- C. Diphenhydramine 50 mg IV alone
- D. No premedication; use gadolinium instead
- E. Epinephrine 0.3 mg IM prophylactically

Q81. A 58-year-old man had DES placement for NSTEMI 6 weeks ago and has tolerated aspirin plus ticagrelor without bleeding. He has no high-bleeding-risk features. Per the 2025 ACC/AHA ACS guideline, what is the Class 1 recommendation now?

- A. Continue aspirin + ticagrelor for 12 months, then stop ticagrelor
- B. Transition to ticagrelor monotherapy, stopping aspirin
- C. Transition to aspirin monotherapy, stopping ticagrelor
- D. De-escalate to aspirin + clopidogrel
- E. Add rivaroxaban 2.5 mg twice daily

Q90. A 66-year-old on warfarin (INR 2.0) needs elective coronary angiography. Per SCAI 2021 consensus, which access is acceptable without further INR reduction?

- A. Femoral, because INR ≤ 2.0 is safe for any access
- B. Radial, because INR ≤ 2.2 is acceptable for radial while femoral requires ≤ 1.8
- C. Neither; give vitamin K and wait
- D. Brachial cutdown
- E. Femoral with a closure device

Q93. A patient with a prior anaphylactoid reaction to contrast develops hypotension, urticaria and wheezing after contrast injection despite premedication. Which is the correct first-line treatment?

- A. Methylprednisolone 125 mg IV
- B. Diphenhydramine 50 mg IV
- C. Epinephrine 0.3–0.5 mg IM (1 mg/mL) into the thigh, or 0.1 mg IV (0.1 mg/mL) slowly if hypotensive
- D. Epinephrine 1 mg IV push
- E. Glucagon 5 mg IV push

Q111. During a prolonged femoral access attempt, 40 mL of 1% lidocaine has been infiltrated in a 60-kg patient, who then develops perioral numbness, tinnitus and a generalized seizure. What is the diagnosis and the specific treatment?

- A. Vasovagal reaction; atropine
- B. Local anesthetic systemic toxicity; airway support, benzodiazepine, and 20% lipid emulsion 1.5 mL/kg bolus
- C. Contrast reaction; epinephrine
- D. Air embolism; 100% oxygen
- E. Cholesterol embolization; supportive care

Q118. A 72-year-old with NSTEMI developed heparin-induced thrombocytopenia (positive serotonin-release assay) during an admission 3 weeks ago; platelets have recovered. She now needs PCI. Which anticoagulation strategy is appropriate?

- A. Unfractionated heparin with an ACT target of 250 s, because antibodies clear after 100 days
- B. Enoxaparin 0.5 mg/kg IV
- C. Bivalirudin bolus and infusion (or argatroban), with no heparin exposure including flushes and heparin-coated devices
- D. Fondaparinux 2.5 mg SC before the procedure
- E. Warfarin to INR 2–3, then proceed

Q119. Two weeks after PCI for NSTEMI, a patient on ticagrelor reports persistent dyspnea with a normal examination, oxygen saturation and chest x-ray; a Holter shows asymptomatic nocturnal ventricular pauses up to 3 seconds. The team decides to switch to clopidogrel. Which statement is correct?

- A. The dyspnea indicates ticagrelor-induced heart failure; stop all P2Y₁₂ inhibition
- B. Both effects relate to ticagrelor's inhibition of adenosine reuptake (ENT-1); switch with a clopidogrel 600 mg load given 24 h after the last ticagrelor dose
- C. Switch to clopidogrel 75 mg daily without a loading dose
- D. Switch to prasugrel 10 mg, which causes the same side effects
- E. The dyspnea resolves only if the ticagrelor dose is reduced to 45 mg

Q120. A patient with STEMI and cardiogenic shock (BP 70/40, cold extremities, lactate 5 mmol/L) is in the cath lab before mechanical support is available. Which first-line vasoactive strategy is best supported by evidence?

- A. Dopamine 10 mcg/kg/min
- B. Norepinephrine titrated to a MAP of at least 65 mmHg, adding dobutamine (or milrinone) for low output
- C. Phenylephrine infusion
- D. Epinephrine as the first-line agent
- E. Vasopressin alone

Q121. You plan to measure FFR using regadenoson. Which statement is correct?

- A. Regadenoson is a non-selective adenosine agonist given as a weight-based infusion
- B. A single 400 mcg IV bolus produces hyperemia within 1–2 minutes lasting several minutes; it is A_{2A}-selective, so bronchospasm is less likely than with adenosine, and aminophylline reverses it
- C. It is unaffected by recent caffeine intake
- D. It is contraindicated in renal failure
- E. It never causes AV block

Imaging

Q7. OCT after stenting shows a signal-poor region with sharply delineated borders and low attenuation behind well-apposed struts. What does this represent?

- A. Lipid pool
- B. Calcium
- C. Red thrombus
- D. Macrophage accumulation
- E. Stent malapposition

Q8. Which patient is best served by IVUS rather than OCT for PCI guidance?

- A. Distal LAD dissection after balloon angioplasty
- B. Ostial left main stenosis in a patient with eGFR 25
- C. In-stent restenosis to characterize neoatherosclerosis
- D. Suspected plaque erosion in a young smoker with NSTEMI
- E. Assessment of stent strut apposition and coverage

Q9. After LM–LAD crossover stenting, IVUS shows a minimum stent area of 5.5 mm² in the distal LM and 5.0 mm² in the ostial LAD. Based on established IVUS criteria, what should be done?

- A. Accept the result; both exceed 4.5 mm²
- B. Post-dilate; distal LM should be ≥ 7 –8 mm² and ostial LAD ≥ 6 mm²
- C. Only the LAD needs post-dilation
- D. Add a second stent into the LCx
- E. Perform FFR of the LM to decide

Q10. Which finding on pre-PCI intravascular imaging most strongly argues for atherectomy or lithotripsy before stenting?

- A. IVUS calcium arc of 90° at one frame
- B. OCT calcium arc $>180^\circ$, thickness >0.5 mm, length >5 mm
- C. Positive remodeling with a lipid-rich plaque
- D. Attenuated plaque without calcium
- E. Echolucent plaque adjacent to the ostium

Q11. A cranial view of the left coronary shows the LAD and diagonals well separated but the LCx overlapped. The proximal LAD appears foreshortened. Which projection best separates the proximal LAD from the diagonals with less foreshortening?

- A. RAO caudal
- B. LAO caudal (spider)
- C. LAO cranial
- D. AP caudal
- E. Lateral

Q114. Which change in fluoroscopic technique increases image contrast for iodinated agents?

- A. Raising tube voltage from 70 to 100 kVp
- B. Operating near 60–80 kVp, close to the iodine K-edge of 33 keV
- C. Removing the copper filter
- D. Increasing the frame rate
- E. Increasing source-to-image distance

Q131. IVUS of an angiographically intermediate (50%) distal left main shows a minimal lumen area of 5.2 mm² in a Western patient. FFR is not available. What is the recommended management?

- A. Defer revascularization; an MLA above 4 mm² is non-ischemic
- B. Revascularize; an MLA below 6 mm² correlates with FFR ≤ 0.80 (Western threshold; about 4.5 mm² in Asian cohorts) and deferral is unsafe
- C. Repeat angiography in 6 months
- D. Revascularize only if plaque burden exceeds 70%
- E. Medical therapy, because left main lesions do not benefit from PCI

Q132. You need to assess the distal RCA bifurcation into the posterior descending and posterolateral branches without overlap. Which projection is best?

- A. RAO 30°
- B. LAO 30–45° with 20–30° cranial angulation
- C. LAO caudal
- D. Left lateral
- E. AP

Physiology and basic science

Q1. A 64-year-old with stable angina has a 60% mid-LAD lesion. FFR is measured with IV adenosine 140 mcg/kg/min through a peripheral IV and reads 0.83 with a modest, unstable Pd/Pa tracing. The patient drank two cups of coffee 2 hours ago. What is the most appropriate next step?

- A. Defer PCI; FFR 0.83 is non-ischemic
- B. Give IC adenosine 200 mcg bolus into the left coronary and repeat
- C. Increase IV adenosine to 180-200 mcg/kg/min and check for a stable plateau
- D. Switch to iFR, which is unaffected by caffeine
- E. Proceed with PCI because the lesion is angiographically significant

Q2. During FFR of an ostial RCA lesion with a 6F JR4 guide, Pa shows ventricularization and the FFR reads 0.86. What is the most likely problem?

- A. Wire drift
- B. Guide catheter damping falsely lowering Pa and raising FFR
- C. Inadequate hyperemia from RCA adenosine-induced AV block
- D. Pressure recovery phenomenon
- E. Nothing; 0.86 is a valid negative result

Q3. Two serial lesions are present in the LAD (proximal 60%, mid 70%). Hyperemic pullback shows a gradient across each. After stenting the proximal lesion, FFR distal to the mid lesion is 0.74. Before the proximal stent it was 0.78 measured at the same distal position. Why did the distal FFR fall?

- A. Distal embolization from the proximal stent
- B. The proximal lesion had limited hyperemic flow, underestimating the mid lesion's severity
- C. Wire drift after stent delivery
- D. Adenosine tachyphylaxis
- E. Stent edge dissection

Q4. Which statement about iFR is correct?

- A. It requires maximal hyperemia with adenosine
- B. The ischemic threshold is ≤ 0.80
- C. It is measured during the diastolic wave-free period when microvascular resistance is stable and minimal
- D. It was inferior to FFR for 1-year MACE in DEFINE-FLAIR
- E. It cannot be used for pullback assessment

Q5. IMR is 45 in the LAD after primary PCI for anterior STEMI with TIMI 3 flow. Which is true?

- A. IMR <25 defines microvascular dysfunction
- B. This IMR predicts poor LV recovery and adverse outcomes
- C. IMR is calculated as Pd multiplied by resting mean transit time
- D. IMR is only valid when FFR is >0.80
- E. An IMR this high mandates intracoronary vasodilators

Q6. In which scenario is FFR most likely to be falsely NEGATIVE (higher than the true value)?

- A. Stable angina with a discrete proximal LAD lesion
- B. Culprit lesion during acute STEMI
- C. Small-vessel diagonal lesion
- D. Left main lesion with a normal LAD and LCx
- E. Chronic total occlusion collateral donor vessel

Q83. Which statement about the 5-year outcomes of the iFR trials is correct?

- A. iFR reduced all-cause mortality compared with FFR in both DEFINE-FLAIR and iFR-SWEDEHEART
- B. DEFINE-FLAIR showed higher all-cause death with iFR (9.0% vs 6.2%), a signal not seen in iFR-SWEDEHEART
- C. Both trials showed higher MACE with iFR at 5 years
- D. iFR was withdrawn from guidelines after the 5-year data
- E. The 5-year data showed more revascularization with iFR

Q129. Which combination of intravascular imaging features best identifies a non-culprit plaque at highest risk of a future event (PROSPECT)?

- A. Calcified plaque with negative remodeling
- B. Plaque burden $\geq 70\%$, minimal lumen area $\leq 4 \text{ mm}^2$, and thin-cap fibroatheroma on VH-IVUS
- C. Fibrous plaque with a lumen area $> 6 \text{ mm}^2$
- D. Spotty calcium alone
- E. Any plaque with $> 50\%$ diameter stenosis

Q130. Angiography shows a 30% stenosis in the proximal LAD, but IVUS at the same site shows a plaque burden of 55% with a vessel area larger than the adjacent reference. Which concept explains the discrepancy?

- A. Angiographic magnification error
- B. Glagov compensatory (positive) remodeling: the vessel enlarges to preserve the lumen until plaque burden reaches about 40%, so angiography underestimates atherosclerosis
- C. Negative remodeling
- D. Vasospasm at the time of angiography
- E. IVUS overestimates plaque because of blooming artifact

Complications

Q22. After post-dilation of a calcified mid-RCA stent with a 3.5 mm NC balloon at 22 atm, the angiogram shows a jet of contrast into the pericardial space. What is the immediate first step?

- A. Give protamine
- B. Perform emergent pericardiocentesis
- C. Reinflate a balloon at the perforation site at low pressure to occlude it
- D. Deliver a covered stent
- E. Call cardiac surgery and stop

Q23. Three hours after uncomplicated PCI of a tortuous circumflex using a hydrophilic polymer-jacketed wire, the patient becomes hypotensive with a JVP of 14 cm. Echo shows a large pericardial effusion. What is the most likely cause?

- A. Retroperitoneal bleed
- B. Acute stent thrombosis
- C. Distal wire perforation of a small branch
- D. Coronary spasm
- E. Aortic dissection

Q24. Post-stent angiography shows an extraluminal cap of contrast at the distal stent edge that persists after the dye clears from the lumen; flow is TIMI 3. What NHLBI dissection type is this and what should be done?

- A. Type A; observe
- B. Type B; observe
- C. Type C; stent
- D. Type D; stent
- E. Type F; emergent surgery

Q25. Which is the most common correctable mechanism of early (<30 day) stent thrombosis found on intravascular imaging?

- A. Hypersensitivity to the polymer
- B. Stent under-expansion
- C. Neoatherosclerosis
- D. Stent fracture
- E. Late acquired malapposition

Q26. A patient becomes hypotensive 40 minutes after femoral sheath removal. Pulse 110, flank pain, no visible groin hematoma. The puncture on the final angiogram was above the inferior epigastric artery origin. What is the best next step?

- A. Atropine and fluids for a vagal reaction
- B. Ultrasound-guided compression of the groin
- C. Non-contrast CT abdomen/pelvis, fluid resuscitation and type-and-cross, with low threshold for contralateral covered stent
- D. Immediate surgical exploration
- E. Observe with serial hemoglobins

Q27. Which statement about radial artery occlusion is correct?

- A. It is best prevented by prolonged occlusive compression
- B. Heparin ≥ 5000 U and patent hemostasis reduce its incidence
- C. It typically causes hand ischemia requiring surgery
- D. The Allen test reliably predicts who will get hand ischemia
- E. Ulnar compression is contraindicated once it occurs

Q28. During rotational atherectomy of a calcified dominant RCA the heart rate falls to 30 with hypotension. What is the mechanism and the preventive measure that should have been taken?

- A. Vagal reaction; atropine before the procedure
- B. Adenosine release from microparticle embolization; prophylactic temporary pacing wire
- C. Perforation; covered stent readiness
- D. Air embolism; aspiration through the guide
- E. Burr entrapment; 2-guide technique

Q29. A patient develops livedo reticularis, blue toes with palpable pulses, eosinophilia and a rising creatinine two weeks after a difficult femoral catheterization. What is the diagnosis?

- A. Contrast-induced nephropathy
- B. Cholesterol embolization syndrome
- C. Heparin-induced thrombocytopenia with thrombosis
- D. Vasculitis from contrast
- E. Femoral artery thrombosis

Q126. While delivering a 3.0×18 mm DES through a tortuous, calcified RCA, the stent strips off the balloon and lies unexpanded in the proximal vessel; the guidewire still passes through its lumen. What is the best initial retrieval strategy?

- A. Push the stent distally with a balloon and deploy it wherever it lands
- B. Advance a 1.5 mm balloon through the stent on the retained wire, inflate it distal to the stent, and withdraw stent and balloon together into the guide
- C. Remove the wire and snare the stent from the aorta
- D. Immediately deploy a second stent to crush it
- E. Refer for surgical removal

Q127. During a left coronary injection the patient develops chest pain, ST elevation and hypotension; angiography shows slow flow with discrete radiolucent columns in the LAD, and an air bubble is seen in the manifold. What is the most appropriate management?

- A. Covered stent for presumed perforation
- B. 100% oxygen, aspiration through the guide or an aspiration catheter, hemodynamic support, and forceful saline injection or balloon dilation to disperse the air if needed
- C. Intracoronary thrombolysis
- D. Immediate CABG
- E. Protamine

Q128. On engaging the left main with an AL2 guide, a contrast stain appears at the LM ostium and extends into the proximal LAD; flow is TIMI 3 and the patient is stable. Which is the correct management?

- A. Withdraw the guide, stop all injections, and observe for 30 minutes
- B. Exchange for a more aggressive guide and inject to define the dissection
- C. Keep the guide position, avoid forceful injections, wire the true lumen with a workhorse wire (IVUS if unsure), and stent the ostial LM entry point; if the dissection extends into the aorta, obtain CT and involve surgery
- D. Emergent CABG without further intervention
- E. Intra-aortic balloon pump and medical therapy

Hemodynamics and mechanical support

Q30. Right heart cath: SVC sat 68%, IVC sat 74%, RA 82%, PA 83%, arterial 97%. Hgb 13, assumed VO_2 250 mL/min. What is the approximate Q_p/Q_s ?

- A. 1.2
- B. 1.5
- C. 2.0
- D. 2.7
- E. 3.4

Q31. A patient with dyspnea has RA 18, RV 42/16, PA 40/18 mean 26, PCWP 18, LVEDP 19, with a dip-and-plateau LV tracing. Simultaneous LV/RV tracings show LV systolic pressure falling while RV systolic pressure rises during inspiration. What is the diagnosis?

- A. Restrictive cardiomyopathy
- B. Constrictive pericarditis
- C. Cardiac tamponade
- D. Severe tricuspid regurgitation
- E. Pulmonary arterial hypertension

Q32. Which hemodynamic finding is most consistent with cardiac tamponade?

- A. Prominent y descent in the RA tracing
- B. Kussmaul sign with steep x and y descents
- C. Blunted/absent y descent with equalization of RA, RV diastolic, PA diastolic and PCWP
- D. LVEDP exceeding RVEDP by 10 mmHg
- E. Pulsus alternans

Q33. A patient in cardiogenic shock after anterior STEMI has MAP 60 mmHg, CO 3.0 L/min, RA 8, PA 30/14, PCWP 24. Calculate cardiac power output and PAPI and interpret.

- A. CPO 0.4 W, PAPI 2.0: LV-dominant shock with poor prognosis
- B. CPO 0.8 W, PAPI 0.5: RV failure
- C. CPO 1.2 W, PAPI 2.0: adequate perfusion
- D. CPO 0.4 W, PAPI 0.5: biventricular failure
- E. CPO 0.6 W, PAPI 4.0: normal

Q34. During a post-PVC beat in a patient with a dynamic LVOT gradient, the aortic pulse pressure decreases while the LV–aortic gradient increases. What is this sign and what does it indicate?

- A. Kussmaul sign; constriction
- B. Brockenbrough–Braunwald–Morrow sign; dynamic obstruction as in HOCM
- C. Pulsus paradoxus; tamponade
- D. Bisferiens pulse; severe aortic regurgitation
- E. Corrigan pulse; fixed aortic stenosis

Q35. Which of the following is a contraindication to intra-aortic balloon pump placement?

- A. Moderate aortic stenosis
- B. Moderate-to-severe aortic regurgitation
- C. Cardiogenic shock after STEMI
- D. Mechanical mitral valve
- E. Atrial fibrillation

Q36. A patient on peripheral VA-ECMO for AMI cardiogenic shock develops pulmonary edema, the aortic valve no longer opens on echo, and spontaneous contrast is seen in the LV. What is the best next step?

- A. Increase ECMO flow
- B. Add an LV venting strategy such as Impella or IABP
- C. Start milrinone
- D. Convert to VV-ECMO
- E. Decrease anticoagulation

Q37. On VA-ECMO with femoral arterial return, the right radial arterial blood gas shows PaO₂ 55 mmHg while the ECMO circuit post-oxygenator PaO₂ is 400 mmHg. What is happening?

- A. Oxygenator failure
- B. Differential hypoxemia (Harlequin syndrome): the recovering native heart ejects poorly oxygenated blood to the upper body
- C. Cannula malposition into the IVC
- D. Recirculation
- E. Hemolysis

Q91. A comatose survivor of out-of-hospital cardiac arrest is admitted after primary PCI. Per TTM2 and the 2025 AHA guideline, what is the recommended temperature management?

- A. Cool to 33 °C for 24 hours
- B. Cool to 32 °C for 48 hours
- C. Deliberate temperature control between 32 and 37.5 °C for at least 36 hours with fever avoidance
- D. No temperature management is needed
- E. Rewarm rapidly to 38 °C

Q92. A 40-year-old with a 22 mm secundum ASD and RV enlargement has PVR 3.5 Wood units on right heart catheterization. According to the 2025 ACHD guideline, what is the appropriate plan?

- A. Close the defect routinely because PVR <5 WU
- B. Do not close; any PVR >2 WU is an absolute contraindication
- C. Closure may be considered at an expert center (PVR 2 to <5 WU), possibly after vasodilator therapy or with a fenestrated device
- D. Refer for lung transplant evaluation
- E. Close only if Qp/Qs is >3

Q136. A 78-year-old has AVA 0.8 cm², mean gradient 24 mmHg, LVEF 30% and stroke volume index 28 mL/m². During dobutamine infusion (up to 20 mcg/kg/min) stroke volume rises 25%, the mean gradient rises to 42 mmHg and the AVA remains 0.85 cm². What is the interpretation?

- A. Pseudo-severe AS; no intervention
- B. True severe low-flow low-gradient AS with contractile reserve; aortic valve replacement (TAVR or SAVR) is indicated
- C. Moderate AS; repeat in 1 year
- D. Absence of contractile reserve; prohibitive risk
- E. Dobutamine testing is contraindicated in this setting

Structural heart

Q38. A 79-year-old with severe symptomatic AS, STS 3%, tricuspid valve, annulus area 420 mm², left coronary height 8 mm with sinus of Valsalva 27 mm, and bulky leaflets is planned for TAVR. Which complication is this anatomy most associated with, and how do you mitigate it?

- A. Annular rupture; downsizing the valve
- B. Coronary obstruction; coronary protection with a wire/undeployed stent or BASILICA
- C. Pacemaker requirement; higher implant
- D. Paravalvular leak; oversizing
- E. Stroke; cerebral embolic protection

Q39. Which is the strongest pre-procedural predictor of permanent pacemaker after TAVR?

- A. Pre-existing left bundle branch block
- B. Pre-existing right bundle branch block
- C. Atrial fibrillation
- D. First-degree AV block alone
- E. Balloon-expandable valve

Q40. After transfemoral TAVR with a balloon-expandable valve, a patient with no other indication for anticoagulation asks about antithrombotic therapy. Based on POPular TAVI, which is preferred?

- A. Aspirin plus clopidogrel for 6 months
- B. Aspirin alone
- C. Rivaroxaban 10 mg plus aspirin
- D. Warfarin for 3 months
- E. Clopidogrel alone

Q41. A 45-year-old woman with rheumatic mitral stenosis (MVA 1.1 cm², mean gradient 12 mmHg, PASP 55) has a Wilkins score of 7, mild MR, and no LA thrombus on TEE. What is the best treatment?

- A. Surgical mitral valve replacement
- B. Percutaneous mitral balloon valvuloplasty
- C. Transcatheter edge-to-edge repair
- D. Medical therapy with beta-blockers and reassess in 1 year
- E. Surgical commissurotomy

Q42. Which patient with secondary mitral regurgitation is most likely to benefit from TEER based on COAPT?

- A. EF 15%, LVESD 75 mm, EROA 20 mm², not on GDMT
- B. EF 30%, LVESD 55 mm, EROA 40 mm², symptomatic on maximally tolerated GDMT
- C. EF 55% with severe primary flail leaflet
- D. EF 25% with mean gradient 8 mmHg across the mitral valve
- E. EF 35% with untreated severe AS

Q43. For transseptal puncture during MitraClip, where should the puncture be located relative to the mitral annulus?

- A. Inferior and anterior, 2 cm above the annulus
- B. Posterior and superior, 4–5 cm above the annulus
- C. Mid-fossa, as low as possible
- D. Anterior near the aorta for maximal working space
- E. Location does not matter with a steerable guide

Q44. A 48-year-old with a cryptogenic cortical stroke has a PFO with a large right-to-left shunt and atrial septal aneurysm; 30-day monitoring shows no AF. What is the recommended management?

- A. Aspirin alone
- B. Warfarin
- C. Percutaneous PFO closure plus antiplatelet therapy
- D. DOAC
- E. PFO closure only if a second stroke occurs

Q45. A 40-year-old has a 24 mm secundum ASD with RV enlargement. Right heart cath: PA 78/34 mean 52, PCWP 10, CO 5 L/min, systemic pressure 118/70 mean 86, PVR 8.4 WU, SVR 15 WU. What is the appropriate management?

- A. Proceed with device closure
- B. Surgical closure
- C. Do not close now; PVR/SVR ratio >2/3 and PAP >2/3 systemic; consider vasodilator testing/PH therapy and reassess
- D. Fenestrated device closure
- E. Balloon test occlusion then device closure regardless of results

Q46. Six months after Amplatzer septal occluder placement for a 30 mm ASD with a deficient retroaortic rim, a patient presents with tamponade. What is the likely mechanism?

- A. Late device embolization
- B. Device erosion of the atrial roof/aorta
- C. Infective endocarditis
- D. Nickel hypersensitivity
- E. Pulmonary vein perforation

Q47. Which post-implant regimen is standard after Watchman FLX in an OAC-eligible patient?

- A. DAPT for 12 months
- B. Warfarin/DOAC plus aspirin for 45 days, TEE, then DAPT to 6 months, then aspirin
- C. Aspirin alone from day 1
- D. Warfarin for 12 months
- E. No antithrombotics

Q48. After alcohol septal ablation for obstructive HCM, which conduction abnormality is most typical, and what is the approximate rate of permanent pacemaker requirement?

- A. LBBB; 2%
- B. RBBB; about 10%
- C. Complete heart block in 40%
- D. No conduction change
- E. Left anterior fascicular block; 25%

Q85. An 82-year-old with severe aortic stenosis is asymptomatic with normal LVEF and a normal exercise test. What does the EARLY TAVR trial and subsequent FDA action support?

- A. Surveillance until symptoms develop is mandatory
- B. TF-TAVR reduced death/stroke/unplanned CV hospitalization vs surveillance, and TAVR is FDA-approved for asymptomatic severe AS
- C. Only SAVR is approved for asymptomatic AS
- D. Balloon valvuloplasty as a bridge
- E. Repeat echo in 2 years

Q86. Which transcatheter valve is FDA-approved for symptomatic severe native aortic regurgitation at high surgical risk?

- A. SAPIEN 3 Ultra RESILIA
- B. Evolut FX+
- C. JenaValve Trilogy
- D. Navitor
- E. Any self-expanding valve with 20% oversizing

Q87. A 76-year-old with severe AS and a 90% proximal LAD lesion is planned for TAVR. Based on NOTION-3 (2024), what is the recommended approach to the coronary lesion?

- A. Never perform PCI before TAVR because ACTIVATION showed harm
- B. PCI of significant lesions (FFR ≤ 0.80 or $\geq 90\%$ stenosis, proximal, ≥ 2.5 mm) reduced death/MI/urgent revascularization at 2 years
- C. PCI only if angina is present
- D. Defer all PCI until after TAVR
- E. Recommend SAVR + CABG regardless of anatomy

Q88. Which statement about the 2026 LAAO trials is correct?

- A. CHAMPION-AF showed Watchman FLX superior to DOAC for ischemic stroke
- B. CLOSURE-AF established LAAO as superior to medical therapy in high-bleeding-risk patients
- C. CHAMPION-AF: non-inferior for CV death/stroke/SE with less non-procedural bleeding but numerically more ischemic stroke; CLOSURE-AF: LAAO failed non-inferiority
- D. Both trials led to a Class 1 recommendation for LAAO in all OAC-eligible AF
- E. Peri-device leak < 5 mm has no prognostic significance

Peripheral and noncardiac vascular

Q65. A 76-year-old with a symptomatic 80% internal carotid stenosis, no prior neck surgery, type II arch, and moderate calcification is being considered for revascularization. Based on CREST, which is preferred?

- A. Transfemoral carotid stenting
- B. Carotid endarterectomy
- C. Medical therapy alone
- D. Transcarotid artery revascularization is contraindicated
- E. Either, with equal periprocedural stroke risk

Q66. Which statement about atherosclerotic renal artery stenosis is correct?

- A. Stenting reduces CV events compared with medical therapy (CORAL)
- B. Stenting should be reserved for flash pulmonary edema, refractory hypertension with bilateral stenosis, or progressive renal failure
- C. Angioplasty alone is adequate for atherosclerotic ostial lesions
- D. Stenting is first-line for all stenoses $> 70\%$
- E. Fibromuscular dysplasia requires stenting

Q67. A patient with intermediate-high-risk PE (RV dilation, troponin positive, SBP 110) is anticoagulated. Based on PEITHO, what is the role of full-dose systemic thrombolysis?

- A. Routine use reduces mortality
- B. It reduces hemodynamic decompensation but increases major and intracranial bleeding without a mortality benefit; reserve for deterioration
- C. It is contraindicated
- D. Half-dose lysis is the standard of care
- E. Catheter-directed therapy is class 1

Q68. Type II endoleak after EVAR is best described as:

- A. Failure of the proximal or distal seal zone
- B. Retrograde flow from lumbar or inferior mesenteric branches into the sac; treat if the sac enlarges
- C. Fabric tear or component separation
- D. Graft porosity
- E. Sac expansion without a visible leak

Q84. A 68-year-old asymptomatic patient has 80% internal carotid stenosis on duplex, confirmed by CTA. Which statement reflects CREST-2 (2025) and current CMS policy?

- A. Neither CAS nor CEA adds to intensive medical management
- B. CAS plus intensive medical management reduced stroke/death vs medical management alone, and CMS covers CAS for asymptomatic $\geq 70\%$ stenosis regardless of surgical risk
- C. Only CEA reduced stroke in the asymptomatic cohort
- D. CMS covers CAS only for high surgical risk
- E. Asymptomatic stenosis should never be revascularized

Q95. Which statement about the 2024 ACC/AHA PAD guideline antithrombotic recommendations is correct?

- A. Aspirin alone for all symptomatic PAD; add nothing after revascularization
- B. Full-dose warfarin plus aspirin after revascularization
- C. Single antiplatelet therapy for symptomatic PAD, and rivaroxaban 2.5 mg BID plus aspirin (Class 1) for symptomatic PAD and after revascularization
- D. Ticagrelor is superior to clopidogrel in PAD (EUCLID)
- E. Vorapaxar is first-line

Q138. A 68-year-old with chronic limb-threatening ischemia (rest pain and a non-healing toe ulcer) has a long superficial femoral artery occlusion and an adequate single-segment great saphenous vein. Based on BEST-CLI, which revascularization is preferred?

- A. Endovascular therapy first in all chronic limb-threatening ischemia
- B. Surgical bypass with the single-segment great saphenous vein, which reduced major adverse limb events or death compared with endovascular therapy; without adequate vein the strategies were equivalent
- C. Primary amputation
- D. Medical therapy with cilostazol
- E. Drug-coated balloon only

Anatomy

Q96. On a left coronary injection in the RAO caudal view, the circumflex is not seen. A right coronary injection shows a vessel arising from the right sinus that passes posterior to the aortic root, appearing as a 'dot' behind the aorta in the RAO projection. What is the diagnosis and its significance?

- A. Anomalous LCx from the right sinus with retroaortic course; benign
- B. Anomalous LCx with interarterial course; high risk of sudden death
- C. Anomalous LM from the right sinus; surgical unroofing
- D. Coronary artery fistula
- E. Left dominance with absent circumflex

Q97. A 24-year-old athlete with exertional syncope has CCTA showing the left main arising from the right sinus with a slit-like ostium and an intramural, interarterial course between the aorta and pulmonary artery. What is the recommended management?

- A. Observation with activity restriction only
- B. Beta-blocker and repeat CT in 1 year
- C. Surgical unroofing (Class 1 with symptoms or ischemia)
- D. PCI with stenting of the intramural segment
- E. Bypass grafting with LIMA

Q98. During a right radial approach the guidewire repeatedly enters the ascending aorta only with difficulty and CT later shows the right subclavian artery arising distal to the left subclavian and passing behind the esophagus. What is this variant and its practical consequence?

- A. Bovine arch; no consequence
- B. Arteria lusoria (aberrant right subclavian, ~1%); right radial access to the coronaries is difficult, so use left radial or femoral
- C. Radial artery loop; straighten with a hydrophilic wire
- D. High-bifurcating radial; use a smaller sheath
- E. Persistent left SVC; affects venous access only

Q99. You need to engage a saphenous vein graft to the right coronary artery, which typically arises from the anterior aortic wall above the native RCA and courses inferiorly. Which catheter is the best first choice?

- A. Left coronary bypass (LCB) catheter
- B. Multipurpose (MP) or JR4
- C. Internal mammary (IM) catheter
- D. Amplatz left 2
- E. Pigtail

Q100. A pulmonary artery catheter advanced from the left internal jugular vein descends along the left heart border and enters a dilated coronary sinus behind the left atrium. What is the most likely anatomy?

- A. Catheter in the azygos vein
- B. Persistent left superior vena cava draining to the coronary sinus
- C. Scimitar syndrome
- D. Unroofed coronary sinus with left-to-right shunt
- E. Catheter perforation into the pericardium

Q101. A 55-year-old woman with a 40% inferior wall infarct has angiography showing the LAD supplying the posterior descending artery via the apex and the RCA ending before the crux, giving no PDA. Which statement is correct?

- A. This is right dominance
- B. This is left dominance: the LCx gives the PDA and posterolateral branches in ~8–10% of people
- C. This is co-dominance
- D. The PDA cannot arise from the LAD
- E. The AV nodal artery must come from the RCA

Q102. An adult presents with heart failure and a continuous murmur; angiography shows a massively dilated RCA with collaterals filling the LAD retrogradely and opacifying the pulmonary artery. What is the diagnosis and treatment?

- A. Coronary artery fistula to the RV; coil embolization
- B. ALCAPA (Bland-White-Garland), adult form; surgical reimplantation of the left coronary
- C. Anomalous LCx from the pulmonary artery; observation
- D. Patent ductus arteriosus; device closure
- E. Ruptured sinus of Valsalva aneurysm; surgery

Q116. Angiography shows diffuse dilation of the RCA to 2× the adjacent reference diameter with sluggish flow and a large thrombus in the dilated segment in a patient with NSTEMI. Which is the best management principle?

- A. Stent the entire ectatic segment with a large DES
- B. Antithrombotic therapy (antiplatelet ± anticoagulation) and thrombus management; avoid stenting ectatic segments; screen for abdominal aortic aneurysm
- C. Covered stent across the whole segment
- D. Emergency CABG
- E. Rotational atherectomy

Radiation safety

Q69. After a 4-hour CTO PCI the reference-point air kerma is 7.2 Gy. What is the appropriate action?

- A. No action; only fluoroscopy time >60 min requires follow-up
- B. Document the dose, counsel the patient about possible skin effects, and arrange skin examination in 2–4 weeks
- C. Admit for observation of radiation sickness
- D. Report as a sentinel event to the Joint Commission
- E. Prescribe topical steroids prophylactically

Q70. Which single change most reduces operator radiation dose during a procedure from the right femoral approach?

- A. Switching from 15 to 7.5 frames per second
- B. Using a steep LAO cranial projection instead of RAO
- C. Lowering the table toward the X-ray tube
- D. Increasing magnification
- E. Reducing collimation

Q71. A pregnant interventional fellow asks about dose limits. Which is correct?

- A. She must stop working in the lab
- B. Fetal dose limit is 5 mSv for the pregnancy and 0.5 mSv per month, measured with a badge under the apron at the waist
- C. Fetal dose limit is 50 mSv per year like the operator limit
- D. Only the collar badge is needed
- E. Lead aprons do not protect the fetus

Q72. Which dose metric best correlates with the risk of deterministic skin injury?

- A. Fluoroscopy time
- B. Kerma-area product (DAP)
- C. Reference-point air kerma
- D. Number of cine frames
- E. Effective dose in mSv

Statistics, quality and regulation

Q73. In a trial the primary endpoint occurred in 8% of the treatment arm and 12% of controls. What is the number needed to treat?

- A. 4
- B. 8
- C. 12
- D. 25
- E. 33

Q74. A non-inferiority trial pre-specifies a margin of 2.5% absolute. The observed difference is 0.8% (95% CI -1.2% to 2.8%) in favor of the control. What is the conclusion?

- A. Non-inferiority established
- B. Non-inferiority NOT established because the upper CI bound exceeds the margin
- C. Superiority of the new treatment
- D. Inferiority proven
- E. The trial is underpowered so nothing can be concluded

Q75. A new biomarker has 90% sensitivity and 80% specificity. In a population with 10% prevalence, which is closest to the positive predictive value?

- A. 90%
- B. 80%
- C. 50%
- D. 33%
- E. 10%

Q76. Which analysis is appropriate to compare time-to-first MACE between two randomized arms while adjusting for covariates?

- A. Chi-square test
- B. Student t-test
- C. Cox proportional hazards regression
- D. Logistic regression
- E. Kaplan–Meier with log-rank only

Q77. In a sham-controlled trial such as ORBITA, blinding of patients and outcome assessors primarily reduces which bias?

- A. Selection bias
- B. Performance and detection (ascertainment) bias, including placebo effect
- C. Lead-time bias
- D. Recall bias
- E. Immortal time bias

Q78. A trial is designed with alpha 0.05 and power 80%. What does the power represent?

- A. Probability of a false positive
- B. Probability of detecting a true difference of the pre-specified size ($1 - \beta$)
- C. Probability that the null hypothesis is true
- D. Probability of a type I error
- E. The confidence interval width

Q109. An NCDR CathPCI report flags your program as an outlier for risk-adjusted acute kidney injury. Which statement is correct about the registry metrics and appropriate-use criteria?

- A. AKI is not a reported CathPCI metric
- B. Outlier status often reflects documentation or abstraction errors, so audit the data before changing practice; NCDR metrics include risk-adjusted mortality, bleeding, AKI, 30-day readmission, discharge medications, radiation dose and AUC
- C. AUC classify PCI of asymptomatic single-vessel disease without ischemia as 'appropriate'
- D. Public reporting has no effect on operator risk aversion
- E. An FFR of 0.85 counts as a severe lesion for AUC purposes

Q112. A new transcatheter therapy costs \$40,000 more than standard care and yields 0.5 additional quality-adjusted life-years. Which statement is correct?

- A. The ICER is \$20,000 per QALY, which is considered low value
- B. The ICER is \$80,000 per QALY, in the intermediate-value range by common US thresholds (\$50,000–\$150,000 per QALY)
- C. The therapy is dominant
- D. The therapy is cost-saving
- E. QALYs cannot be compared across diseases

Q113. A patient dies of tamponade after a coronary perforation attributed to a new scoring balloon. Which reporting obligation applies to the hospital?

- A. No obligation; reporting is voluntary through MedWatch
- B. Report the device-related death to both the FDA and the manufacturer within 10 working days under the Safe Medical Devices Act
- C. Report to the state medical board within 30 days
- D. Report only to the NCDR registry
- E. Report to the manufacturer only if the device is Class III

Q137. In a non-inferiority trial comparing a new antithrombotic regimen with standard therapy, which statement about the analysis is correct?

- A. Intention-to-treat analysis is the most conservative, as in superiority trials
- B. Per-protocol (or as-treated) analysis should be reported alongside ITT because crossover and non-adherence dilute differences in ITT and bias toward non-inferiority
- C. A 95% confidence interval that includes zero proves non-inferiority
- D. The non-inferiority margin is chosen after unblinding
- E. Non-inferiority can be claimed when the superiority p-value exceeds 0.05

Part II. Answer key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	31	B	61	C	91	C	121	B
2	B	32	C	62	B	92	C	122	B
3	B	33	A	63	B	93	C	123	C
4	C	34	B	64	C	94	B	124	C
5	B	35	B	65	B	95	C	125	B
6	B	36	B	66	B	96	A	126	B
7	B	37	B	67	B	97	C	127	B
8	B	38	B	68	B	98	B	128	C
9	B	39	B	69	B	99	B	129	B
10	B	40	B	70	A	100	B	130	B
11	C	41	B	71	B	101	B	131	B
12	C	42	B	72	C	102	B	132	B
13	B	43	B	73	D	103	B	133	B
14	C	44	C	74	B	104	B	134	B
15	B	45	C	75	D	105	C	135	B
16	B	46	B	76	C	106	B	136	B
17	B	47	B	77	B	107	B	137	B
18	C	48	B	78	B	108	C	138	B
19	C	49	B	79	B	109	B		
20	C	50	B	80	B	110	C		
21	B	51	C	81	B	111	B		
22	C	52	B	82	B	112	B		
23	C	53	C	83	B	113	B		
24	C	54	B	84	B	114	B		
25	B	55	B	85	B	115	C		
26	C	56	B	86	C	116	B		
27	B	57	B	87	B	117	B		
28	B	58	C	88	C	118	C		
29	B	59	B	89	C	119	B		
30	C	60	C	90	B	120	B		

Part III. Answers and explanations

Case selection

Q49. B. Culprit-only PCI with staged evaluation of the non-culprit lesions

CULPRIT-SHOCK: immediate multivessel PCI increased death or renal replacement therapy at 30 days compared with culprit-only PCI (55.4% vs 45.9%). Complete revascularization (COMPLETE) applies to hemodynamically stable STEMI.

Q50. B. CABG

FREEDOM, SYNTAX and BARI 2D show CABG reduces death/MI versus PCI in diabetics with multivessel disease, especially with intermediate-high SYNTAX scores. FAME 3 showed FFR-guided PCI failed non-inferiority to CABG in 3VD.

Q51. C. Medical therapy; routine PCI of a totally occluded infarct artery >24 h in stable asymptomatic patients has no benefit

OAT: PCI of a persistently occluded infarct artery 3–28 days after MI in stable asymptomatic patients did not reduce death/MI/HF; ACC/AHA gives routine PCI of an occluded infarct artery >24 h in stable asymptomatic patients a Class 3 (no benefit). Between 12 and 24 h, primary PCI is indicated for ongoing ischemia, instability, HF or arrhythmia (Class 1) and is otherwise not routinely recommended in US guidelines; note that ESC allows a routine primary PCI strategy to be considered up to 48 h in asymptomatic patients, so US-style stems signal the intended answer with cues like 'pain resolved', 'stable', 'Q waves' and a time beyond 24 h.

Q52. B. Within 24 h

High-risk features (NSTEMI, dynamic ST, GRACE >140) warrant an early invasive strategy within 24 h (TIMACS, VERDICT). Immediate (<2 h) is reserved for refractory angina, instability, life-threatening arrhythmias, mechanical complications or acute HF.

Q53. C. Conservative management with beta-blocker and aspirin, and screen for fibromuscular dysplasia

This is type 2 spontaneous coronary artery dissection. With preserved flow and no ongoing ischemia, conservative therapy is preferred (spontaneous healing >90%); PCI risks propagation. Beta-blockers reduce recurrence; screen for FMD with brain-to-pelvis imaging.

Q54. B. Revascularization does not reduce death or MI overall but improves angina and quality of life

ISCHEMIA: an initial invasive strategy did not reduce death or MI (median 3.2 y) in stable patients with moderate-severe ischemia and no LM disease; it improved angina symptoms, most in those with frequent angina. Proximal LAD disease in MVD may favor CABG for survival (older data).

Q55. B. PCI with DES is a reasonable alternative to CABG (class 2a) with equivalent mortality in low-intermediate SYNTAX

EXCEL, NOBLE, PRECOMBAT, SYNTAX LM subgroup and meta-analyses: no mortality difference at 5 years for low-intermediate SYNTAX, with more spontaneous MI/revascularization after PCI and more stroke early after CABG. Guidelines: PCI 2a for low/intermediate complexity; CABG preferred for high SYNTAX.

Q79. B. Aspirin + clopidogrel + apixaban for ≤1 week, then clopidogrel + apixaban for 6–12 months, then apixaban alone

HBR with OAC: minimize triple therapy (up to 1 week), clopidogrel as the P2Y12 inhibitor with the DOAC for 6 (stable CAD) to 12 (ACS) months, then OAC alone (AFIRE). MASTER DAPT supports abbreviated DAPT in HBR.

Q82. B. No benefit, with higher cardiovascular death in the Impella arm

CHIP-BCIS3 randomized complex PCI in LVEF ≤35% to Impella CP vs no planned MCS and found no benefit, with higher CV death with Impella (26.7% vs 14.5%). Together with PROTECT II (neutral primary endpoint) and BCIS-1 (IABP neutral), routine MCS for high-risk PCI is not supported; PROTECT IV is pending.

Q94. B. Immediate multivessel PCI was non-inferior to staged PCI, with fewer MIs and unplanned revascularizations

MULTISTARS AMI (2023): immediate complete revascularization was non-inferior to staged (19–45 days), with fewer MIs and unplanned revascularizations. BIOVASC was similar. Complete revascularization is Class 1 in stable STEMI; the 2025 guideline slightly prefers a single procedure, with staging within 45 days acceptable. In shock, culprit-only PCI at the index procedure (CULPRIT-SHOCK).

Q103. B. Posteromedial papillary muscle rupture; afterload reduction, IABP/Impella bridge and emergent surgery

Papillary muscle rupture involves the posteromedial muscle (single blood supply from the PDA) after inferior MI, presents with abrupt pulmonary edema and shock, often a soft or absent murmur, giant v waves and no step-up. VSR gives a harsh holosystolic murmur with an RA-to-RV/PA oxygen step-up. Both are surgical emergencies; nitroprusside if BP allows and MCS as a bridge.

Q104. B. Ventricular septal rupture at the apex; highest shock mortality; IABP/afterload reduction and surgery, with device closure for residual or as a bridge

An oxygen step-up from RA to PA (≥5% at the ventricular/PA level) diagnoses ventricular septal rupture. Anterior MI ruptures the apical septum; inferior MI ruptures the basal septum (harder to repair). Risk factors: older age, female, first MI, late presentation, no prior smoking. Mortality in shock is ~87%. Impella carries risk of shunt reversal/embolization in VSR.

Q105. C. Cautious volume loading, maintain AV synchrony, dobutamine if no response, and consider RV mechanical support

RV infarction: preload-dependent, so avoid nitrates and diuretics, give fluids cautiously (RA/PCWP >0.8 confirms RV predominance), preserve AV synchrony (pace if heart block), add dobutamine, and escalate to Impella RP/ProtekDuo if refractory. Bezold–Jarisch bradycardia/hypotension with reperfusion responds to atropine and fluids.

Q106. B. Coronary CT angiography (with CT-FFR for 40–90% lesions if needed)

For intermediate-high risk symptomatic patients with no known CAD, CCTA is favored under 65 (and when not yet on optimal therapy) and stress imaging is favored at 65 and older or with known CAD burden; PET is preferred over SPECT. Low-risk patients can defer testing or use CAC/exercise ECG. Go straight to invasive angiography for CCTA showing ≥50% LM or ≥70% three-vessel disease with symptoms.

Q107. B. Fibrinolysis within 30 minutes of arrival, then transfer to a PCI center with angiography within 2–24 hours

If first medical contact to device will exceed 120 minutes, fibrinolysis within 30 minutes of arrival is Class 1 (symptoms ≤ 12 h), followed by transfer and routine angiography within 2–24 hours (pharmacoinvasive: TRANSFER-AMI, STREAM). Failed lysis ($<50\%$ ST resolution at 60–90 min, ongoing pain, instability) mandates immediate rescue PCI. Facilitated PCI immediately after full-dose lytic is harmful; STREAM halves tenecteplase in patients ≥ 75 .

Q108. C. Selective (ischemia-guided) approach; early invasive evaluation is Class 3 for low likelihood of ACS

Low-risk patients (negative serial troponins, no ST change) get a selective invasive approach with non-invasive testing as indicated; routine early invasive evaluation in those with low ACS likelihood or prohibitive comorbidity is Class 3. Immediate (<2 h) is for instability; early (<24 h) for GRACE >140 , dynamic ST or troponin rise.

Q110. C. Proceed after 3 months is acceptable for time-sensitive surgery; continue aspirin, hold clopidogrel 5 days, resume within 24–48 h

After DES for chronic coronary syndrome, elective surgery is ideally delayed 6 months, but time-sensitive surgery may proceed after 3 months when the risk of delay exceeds the stent-thrombosis risk (after ACS the corresponding intervals are 12 and 6 months). Continue aspirin throughout if the surgeon permits, interrupt the P2Y₁₂ inhibitor for the minimum time, resume promptly, and never interrupt DAPT within 30 days of DES. Heparin bridging does not protect against stent thrombosis; cangrelor bridging is reserved for very high risk. CARP: prophylactic revascularization before surgery does not improve outcomes.

Q115. C. Prophylactic temporary transvenous pacing because the block is infranodal with an unreliable escape

In anterior MI, new bifascicular block with Mobitz II reflects infranodal (His–Purkinje) injury from a large septal infarct; the escape is wide, slow and unreliable, so temporary transvenous pacing is indicated (transcutaneous as a bridge). Inferior MI block is AV nodal, usually transient, with a narrow escape, and responds to atropine. Permanent pacing is decided after the acute phase if block persists.

Q122. B. An initial invasive strategy does not reduce death or MI over about 3.5 years but improves angina control; medical therapy with revascularization for refractory symptoms is appropriate

ISCHEMIA (n=5179): no difference in CV death/MI/UA/HF/cardiac arrest between initial invasive and conservative strategies; invasive management produced more periprocedural MI, fewer spontaneous MI, and better angina-related quality of life in symptomatic patients. Exclusions: LM $\geq 50\%$, EF $<35\%$, recent ACS, unacceptable angina. ISCHEMIA-EXTEND (7 years): lower CV death but higher non-CV death, no all-cause difference.

Q123. C. No reduction in death or heart-failure hospitalization and no sustained improvement in LVEF or quality of life compared with medical therapy alone

REVIVED-BCIS2 (700 patients, EF $\leq 35\%$, extensive CAD, viability in ≥ 4 segments): PCI did not reduce death or HF hospitalization (37.2% vs 38.0%) or improve EF, and the early quality-of-life benefit disappeared by 24 months. STICH (10 years) showed CABG reduced all-cause death versus medical therapy in ischemic cardiomyopathy, and viability did not predict benefit. In HFrEF, CABG is preferred for suitable multivessel disease; PCI is for angina or when surgery is not an option.

Q124. C. Ischemic stroke 6 weeks ago

Absolute contraindications: any prior intracranial hemorrhage, ischemic stroke within 3 months, known structural cerebrovascular lesion or intracranial malignancy, active bleeding (not menses), suspected aortic dissection, significant closed-head or facial trauma within 3 months, intracranial or spinal surgery within 2 months, severe uncontrolled hypertension unresponsive to therapy, and prior streptokinase exposure (for streptokinase).

Relative: BP >180/110, ischemic stroke >3 months, dementia, prolonged CPR, major surgery <3 weeks, recent internal bleeding, pregnancy, active peptic ulcer, oral anticoagulation.

Q125. B. CABG with a LIMA to the LAD (class 1 over PCI in diabetic multivessel disease)

FREEDOM: in diabetics with multivessel disease, CABG reduced death/MI/stroke at 5 years (18.7% vs 26.6%) and all-cause death, with more early stroke. The 2021 revascularization guideline gives CABG a class 1 recommendation over PCI in diabetics with multivessel disease involving the LAD; PCI is reasonable (2a) when the patient is a poor surgical candidate or the anatomy is simple.

Procedural technique

Q56. B. Proximal optimization technique, re-wire the diagonal through a distal cell and perform kissing balloon inflation

Provisional strategy: POT first (opens the stent cell at the carina), re-wire the SB through a distal cell, then kissing balloon (or sequential POT–side–POT). A second stent is reserved for dissection, TIMI <3, or an FFR-positive/large SB after ballooning.

Q57. B. DK crush two-stent technique

DKCRUSH-V: in true distal LM bifurcations, DK crush reduced target lesion failure vs provisional (5.0% vs 10.7% at 1 y). DEFINITION criteria (SB lesion ≥ 10 mm, SB ≥ 2.5 mm) identify complex bifurcations that benefit from planned two-stent strategies.

Q58. C. 3, very difficult

J-CTO points: blunt stump (1), calcification (1), bending $>45^\circ$ (0), length ≥ 20 mm (1), prior failed attempt (0) = 3
→ very difficult (low probability of guidewire crossing within 30 minutes).

Q59. B. Epicardial collaterals; never dilate

Epicardial collateral perforation bleeds into the pericardium (loculated in post-CABG patients); they must never be dilated. Septal collaterals can be dilated with a small (1.5 mm) balloon if needed; septal perforation usually causes a contained septal hematoma.

Q60. C. Avoid decelerations >5000 rpm and use short pecking runs of 15–20 s

Burr-to-artery 0.5–0.6, 140,000–150,000 rpm (lower for larger burrs), short runs, no decelerations >5000 rpm (heat, entrapment, slow-flow), dedicated 0.009 inch RotaWire, flush with heparin/vasodilator. Contraindicated with thrombus, SVG, and dissection.

Q61. C. Sonic pressure waves fracture calcium with a 1:1 sized balloon inflated at 4 atm, ~10 pulses per cycle

IVL uses lithotripsy emitters in a balloon inflated to 4 atm (then 6 atm), delivering 10 pulses per cycle (80 pulses per catheter for C², 120 for C²⁺). It works on deep, eccentric and nodular calcium and causes minimal slow-flow. Ventricular capture ('shocktopics') is benign; pacemakers are not a contraindication.

Q62. B. Distal embolic protection device

SAFER: distal occlusion embolic protection reduced 30-day MACE by 42% in SVG PCI. Direct stenting, slight undersizing and IC vasodilators help; covered stents and BMS have no benefit (SYMBIOT, DIVA).

Q63. B. Paclitaxel drug-coated balloon

AGENT IDE (2024): paclitaxel DCB reduced target lesion failure vs uncoated balloon for coronary ISR, leading to FDA approval. DES re-stenting is an alternative (RIBS); brachytherapy is reserved for multiple recurrences. Atherectomy is for calcified neointima or undilatable stents (IVL preferred).

Q64. C. Sequential techniques (DK crush, culotte) are feasible through 6F (ID ~0.070 inch); simultaneous kissing stents require 7F or larger

6F guides have an inner diameter of ~0.070 inch, allowing a stent plus balloon but not two stents simultaneously. Sequential two-stent techniques are feasible in 6F; simultaneous kissing stents/V-stenting need $\geq 7F$. Burr limits: 1.5 mm (6F), 1.75 (7F), 2.0 (8F).

Q80. B. Polymer-jacketed tapered wire (e.g., Fielder XT, Fighter)

Tapered, polymer-jacketed low-tip-load wires (0.8–1 g) track microchannels for soft/tapered CTOs. Escalate to intermediate stiff tapered (Gaia) then high-tip-load (Confianza Pro 12, Hornet) for hard caps; de-escalate after cap penetration.

Q89. C. Class 2b; neither showed LVEF improvement

CTO PCI for refractory angina after treating non-CTO lesions is Class 2b (2021). EuroCTO showed better angina/QoL; DECISION-CTO showed no MACE difference (with high crossover); EXPLORE and REVASC showed no LVEF benefit. Stop a CTO procedure at air kerma >5 Gy, contrast $>3.7\times$ eGFR, or >3 hours.

Q117. B. A pre-existing LBBB is a reason to have temporary pacing available because biopsy can cause RBBB and complete heart block

Biopsy targets the RV septum (free wall risks perforation), takes 4–5 samples, and is indicated for transplant rejection surveillance, fulminant or unexplained heart failure (giant cell and eosinophilic myocarditis change management), and infiltrative disease. Complications: perforation/tamponade 0.5–1%, tricuspid injury, arrhythmia, and RBBB, which in a patient with LBBB produces complete heart block.

Q133. B. Use a guide-extension catheter advanced over an anchor balloon or balloon shaft, add a buddy wire, and consider a more supportive shape (AL1) or 7F guide after further lesion preparation

Support options: a more supportive guide shape (Amplatz, XB/EBU) or larger size (7F), deep intubation (dissection risk), a buddy wire, an anchor balloon in a side branch, a guide-extension catheter (5-in-6 or 6-in-7), and better lesion preparation (NC balloon, scoring, IVL or atherectomy). Guide extensions risk proximal dissection, stent stripping at the collar and pressure damping, so never advance one unsupported into a diseased proximal segment.

Q134. B. Disengage the guide slightly, position the stent in a view that opens the ostium (LAO or lateral) so it protrudes 1–2 mm into the aorta, deploy, then flare the proximal struts with a short larger balloon partly in the aorta; the Szabo anchor-wire technique is an alternative

Aorto-ostial stenting: first confirm true ostial disease rather than spasm or catheter-induced damping (nitroglycerin, non-selective cusp injection), prepare calcified ostia, pull the guide back during deployment, allow 1–2 mm of protrusion into the aorta, flare the proximal edge with a short NC balloon, and re-engage the guide gently over the wire so it does not catch struts. The Szabo technique anchors the stent at the ostium with a second wire passed through the most proximal strut.

Q135. B. Proximal optimization (POT) and leave the diagonal alone; rewire and dilate only for TIMI flow <3, dissection, or ischemia, or for a large side branch

Provisional strategy: a jailed wire protects the side branch, then POT with a balloon sized to the proximal main vessel; routine final kissing adds no benefit (Nordic III). Intervene on the side branch only when it is compromised (flow <3, ostial stenosis $\geq 75\%$ with symptoms or ischemia, dissection) or large (≥ 2.5 mm): rewire through the distal cell, kiss, then re-POT. Most ostial pinching is carina shift and improves with time.

Pharmacology

Q12. C. Prasugrel 60 mg after coronary anatomy is defined and PCI is chosen

ACCOAST showed that prasugrel pretreatment in NSTEMI increases bleeding without ischemic benefit. ISAR-REACT 5 showed prasugrel (given after anatomy in NSTEMI-ACS) beat ticagrelor (given at presentation) for death/MI/stroke without excess bleeding.

Q13. B. Give ticagrelor 180 mg during the infusion or immediately after it stops

Cangrelor occupies the P2Y₁₂ receptor and blocks the irreversible binding of the active metabolites of clopidogrel/prasugrel, which must be given immediately after the infusion stops. Ticagrelor binds a different site reversibly and can be given during the infusion.

Q14. C. Ticagrelor 180 mg then 90 mg twice daily

Prasugrel showed no net benefit in patients ≥ 75 years or < 60 kg (the 5 mg dose is a labeling compromise without outcome data) and is contraindicated after stroke/TIA. Ticagrelor at standard dosing is preferred; the 60 mg dose is for long-term therapy after year 1 (PEGASUS).

Q15. B. Continuing the full PCI-dose infusion for 2–4 hours after the procedure

Bivalirudin's 25-minute half-life leaves a gap before oral P2Y₁₂ onset in STEMI. BRIGHT, MATRIX and BRIGHT-4 support a post-PCI high-dose infusion (1.75 mg/kg/h for 2–4 h) to prevent acute stent thrombosis.

Q16. B. Eptifibatide

Eptifibatide is renally cleared and contraindicated in dialysis patients; it is dose-reduced (1 mcg/kg/min) for CrCl < 50 . Tirofiban infusion is halved for CrCl ≤ 60 . Abciximab (no longer marketed) was not renally cleared.

Q17. B. Apixaban + clopidogrel, aspirin stopped

AUGUSTUS (2 $\times$ 2): apixaban caused less bleeding than warfarin, and aspirin added bleeding without ischemic benefit. Default: DOAC + clopidogrel with aspirin stopped at discharge (up to 1 week; ≤ 30 days only if very high ischemic risk). Potent P2Y₁₂ agents are avoided in combination with OAC.

Q18. C. Give protamine after the covered stent has sealed the perforation, if bleeding persists

Reversing heparin while wires/balloons are in the vessel and before definitive sealing risks thrombosis of the vessel and equipment. Once the perforation is sealed (covered stent) and if bleeding/tamponade continues, protamine (1 mg/100 U of heparin given in the prior 2–3 h, max ~ 50 mg, slowly) is appropriate. Prior NPH insulin, fish allergy and vasectomy increase reaction risk but are not absolute contraindications.

Q19. C. Limiting contrast volume and isotonic saline hydration guided by LVEDP

PRESERVE showed no benefit of NAC or bicarbonate. POSEIDON supported LVEDP-guided isotonic hydration; contrast volume (volume/CrCl ratio) is the modifiable risk factor. Iso- vs low-osmolar agents show no consistent difference; prophylactic dialysis is harmful/no benefit.

Q20. C. Intracoronary nitroglycerin

No-reflow is microvascular; nitroglycerin dilates epicardial vessels and does not improve microvascular flow, and it worsens hypotension. Adenosine, nicardipine, verapamil and nitroprusside act on the microcirculation and should be delivered distally.

Q21. B. Hydrocortisone 200 mg IV immediately and every 4 hours until the procedure, plus diphenhydramine 50 mg IV

Elective regimen: prednisone 50 mg PO at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h. Emergent regimen: hydrocortisone 200 mg IV immediately and every 4 h until completed plus diphenhydramine. Steroids need several hours to act; gadolinium is not an alternative for coronary angiography.

Q81. B. Transition to ticagrelor monotherapy, stopping aspirin

The 2025 ACS guideline gives a Class 1 recommendation to transition to ticagrelor monotherapy at or after 1 month post-PCI when DAPT has been tolerated (TWILIGHT, TICO, ULTIMATE-DAPT, T-PASS). Step-down now favors P2Y12 monotherapy, not aspirin. De-escalation to clopidogrel after 1 month is 2b, and STOPDAPT-2 ACS showed early clopidogrel monotherapy was not non-inferior in ACS.

Q90. B. Radial, because INR ≤ 2.2 is acceptable for radial while femoral requires ≤ 1.8

SCAI 2021: INR ≤ 2.2 is acceptable for radial access and ≤ 1.8 for femoral; vitamin K is not given. DOACs need no bridging; apixaban/rivaroxaban are held 24–48 h and dabigatran 48 h (longer with CrCl < 50) for femoral procedures. In emergencies, go radial.

Q93. C. Epinephrine 0.3–0.5 mg IM (1 mg/mL) into the thigh, or 0.1 mg IV (0.1 mg/mL) slowly if hypotensive

Epinephrine is the only first-line drug: IM 0.3–0.5 mg of 1 mg/mL (repeat every 5–15 min) or 0.1 mg of 0.1 mg/mL IV slowly (or 10 mcg boluses then 1–4 mcg/min). Never give 1 mg IV push (the arrest dose). Steroids and antihistamines are adjuncts only; glucagon 1–5 mg IV over 5 min is for patients on beta-blockers who are refractory.

Q111. B. Local anesthetic systemic toxicity; airway support, benzodiazepine, and 20% lipid emulsion 1.5 mL/kg bolus

1% lidocaine is 10 mg/mL, so 40 mL = 400 mg, exceeding the plain-lidocaine maximum of 4.5 mg/kg (270 mg here; 7 mg/kg with epinephrine). Toxicity progresses from perioral numbness, tinnitus and agitation to seizures and then cardiovascular collapse. Treat with airway management, benzodiazepines for seizures, and intravenous 20% lipid emulsion (1.5 mL/kg bolus, then 0.25 mL/kg/min); avoid vasopressin and use low-dose epinephrine if needed.

Q118. C. Bivalirudin bolus and infusion (or argatroban), with no heparin exposure including flushes and heparin-coated devices

Any heparin (UFH, LMWH, flushes, coated catheters) is contraindicated in acute or subacute HIT; bivalirudin is the PCI anticoagulant of choice (class 1) and argatroban an alternative (hepatic clearance, prolongs INR). Fondaparinux does not cause HIT but is not an adequate sole PCI anticoagulant (catheter thrombosis in OASIS-5). Re-exposure to UFH is considered only when antibodies are undetectable (typically > 100 days) and no alternative exists.

Q119. B. Both effects relate to ticagrelor's inhibition of adenosine reuptake (ENT-1); switch with a clopidogrel 600 mg load given 24 h after the last ticagrelor dose

Ticagrelor blocks adenosine reuptake through ENT-1 and raises plasma adenosine: dyspnea (~14%, usually early and self-limited) and ventricular pauses (mostly asymptomatic, first week) follow. Because ticagrelor binds reversibly and receptors remain occupied, de-escalation uses a clopidogrel 600 mg load 24 h after the last ticagrelor dose (prasugrel: 60 mg load at 24 h). Escalation from clopidogrel to ticagrelor or prasugrel needs no delay.

Q120. B. Norepinephrine titrated to a MAP of at least 65 mmHg, adding dobutamine (or milrinone) for low output

SOAP II (cardiogenic shock subgroup) found more arrhythmias and higher mortality with dopamine than norepinephrine; OptimaCC found epinephrine caused more refractory shock and lactic acidosis than norepinephrine. Norepinephrine is the preferred first-line vasopressor, with dobutamine or milrinone added for inotropy; phenylephrine is a pure alpha agonist that raises afterload without inotropy.

Q121. B. A single 400 mcg IV bolus produces hyperemia within 1–2 minutes lasting several minutes; it is A2A-selective, so bronchospasm is less likely than with adenosine, and aminophylline reverses it

Regadenoson is an A2A-selective agonist: fixed 400 mcg IV bolus, peak hyperemia within 1–2 min, duration several minutes, less bronchoconstriction (A2B/A3 mediated) than adenosine though caution persists in active wheezing. Caffeine blunts the response and aminophylline 50–250 mg IV reverses it. AV block, flushing and headache occur; renal failure prolongs the effect but is not a contraindication.

Imaging

Q7. B. Calcium

Calcium on OCT is signal-poor with sharp borders and low attenuation; lipid is signal-poor with diffuse borders and high attenuation; red thrombus is highly attenuating/backscattering; macrophages appear as punctate bright spots.

Q8. B. Ostial left main stenosis in a patient with eGFR 25

OCT requires contrast clearance and cannot image the aorto-ostial LM well; renal failure is a further reason to avoid contrast. IVUS penetrates deeper and needs no contrast. OCT excels at dissection, thrombus/erosion, ISR mechanism and strut-level detail.

Q9. B. Post-dilate; distal LM should be $\geq 7\text{--}8\text{ mm}^2$ and ostial LAD $\geq 6\text{ mm}^2$

Kang criteria for LM stenting: MSA $\geq 8.2\text{ mm}^2$ proximal LM, $\geq 7.2\text{ mm}^2$ polygon of confluence, $\geq 6.3\text{ mm}^2$ ostial LAD, $\geq 5.0\text{ mm}^2$ ostial LCx (roughly 8/7/6/5). Under-expansion is the strongest predictor of LM stent failure; post-dilate with NC balloons and POT.

Q10. B. OCT calcium arc $>180^\circ$, thickness $>0.5\text{ mm}$, length $>5\text{ mm}$

The OCT calcium score (arc $>180^\circ = 2$, thickness $>0.5\text{ mm} = 1$, length $>5\text{ mm} = 1$; score 4) predicts stent under-expansion. IVUS arc $>270^\circ$ with length $>5\text{ mm}$ similarly. Attenuated plaque predicts no-reflow, not under-expansion.

Q11. C. LAO cranial

LAO cranial opens the proximal-mid LAD/diagonal bifurcation and separates diagonals. RAO/AP caudal and LAO caudal (spider) are for the LM bifurcation and circumflex system; RAO cranial shows mid/distal LAD.

Q114. B. Operating near 60–80 kVp, close to the iodine K-edge of 33 keV

Iodine's K-edge is 33.2 keV; photoelectric absorption and therefore iodine contrast is maximal with tube voltages in the 60–80 kVp range. Higher kVp improves penetration in large patients but reduces contrast. Copper filtration removes low-energy photons that add dose without image information. Frame rate and magnification change dose, not contrast.

Q131. B. Revascularize; an MLA below 6 mm² correlates with FFR ≤0.80 (Western threshold; about 4.5 mm² in Asian cohorts) and deferral is unsafe

For the left main, MLA <6 mm² corresponds to FFR <0.80 in Western populations (LITRO safely deferred MLA ≥6 mm²); Asian studies use 4.5–4.8 mm². Non-LM MLA thresholds (about 2.4–3 mm²) are poorly specific and physiology should decide. Post-PCI optimization targets: MSA >5.5 mm² (IVUS) or >4.5 mm² (OCT) for non-LM lesions; for the LM, ostial LM ≥8, polygon of confluence ≥7, ostial LAD ≥6 and ostial LCx ≥5 mm².

Q132. B. LAO 30–45° with 20–30° cranial angulation

LAO cranial opens the crux and separates the PDA from the posterolateral branches; RAO shows the mid RCA with little foreshortening and the PDA along its length; straight LAO shows the proximal and mid RCA; the left lateral view displays the RCA ostium and mid vessel and is also useful for the proximal LAD. Conus and SA nodal branches are best seen in RAO.

Physiology and basic science

Q1. C. Increase IV adenosine to 180-200 mcg/kg/min and check for a stable plateau

Caffeine antagonizes adenosine receptors and can blunt hyperemia, giving a falsely high FFR. Peripheral IV adenosine is also less reliable than central. Before deferring, ensure maximal hyperemia: increase the adenosine dose (or use IC adenosine at adequate dose, or regadenoson). IC adenosine is also caffeine-sensitive. iFR is a resting index and is an option, but the question tests recognizing inadequate hyperemia.

Q2. B. Guide catheter damping falsely lowering Pa and raising FFR

Ostial lesions with an engaged guide produce damping/ventricularization: Pa is falsely low, so Pd/Pa is falsely high. Disengage the guide (and never use side-hole catheters for pressure measurement) before recording FFR.

Q3. B. The proximal lesion had limited hyperemic flow, underestimating the mid lesion's severity

In tandem lesions each stenosis limits flow across the other; the true severity of the remaining lesion is unmasked only after the other is treated. Individual FFRs are not additive. Re-measure after each intervention.

Q4. C. It is measured during the diastolic wave-free period when microvascular resistance is stable and minimal

iFR is a resting index measured in the diastolic wave-free period (cutoff ≤0.89). DEFINE-FLAIR and iFR-SWEDEHEART showed non-inferiority to FFR. Co-registered pullback is a strength of iFR.

Q5. B. This IMR predicts poor LV recovery and adverse outcomes

IMR = distal pressure × hyperemic mean transit time; normal <25. Post-STEMI IMR >40 predicts poor recovery/mortality. It is independent of epicardial stenosis when corrected for collateral flow.

Q6. B. Culprit lesion during acute STEMI

Microvascular stunning and obstruction after MI limit hyperemia, so the pressure gradient is smaller and FFR is falsely high in the culprit vessel. FFR of non-culprit vessels is reasonably reliable. Severe LVH and microvascular disease behave similarly.

Q83. B. DEFINE-FLAIR showed higher all-cause death with iFR (9.0% vs 6.2%), a signal not seen in iFR-SWEDEHEART

At 1 year, iFR was non-inferior to FFR with fewer revascularizations and less discomfort. At 5 years, DEFINE-FLAIR reported an unexplained excess of all-cause death with iFR (9.0% vs 6.2%, HR 1.56) despite similar MACE; iFR-SWEDEHEART showed no difference. Resting indices remain guideline-endorsed (Class 1 for intermediate stenosis without prior ischemia testing).

Q129. B. Plaque burden $\geq 70\%$, minimal lumen area $\leq 4 \text{ mm}^2$, and thin-cap fibroatheroma on VH-IVUS

PROSPECT (three-vessel VH-IVUS after ACS): non-culprit events arose from angiographically mild lesions with plaque burden $\geq 70\%$ (HR 5.0), MLA $\leq 4.0 \text{ mm}^2$ (HR 3.2) and VH-TCFA (HR 3.4); all three together carried an 18% three-year event rate. A TCFA has a fibrous cap $< 65 \mu\text{m}$ over a large necrotic core with macrophages and positive remodeling; OCT measures cap thickness directly. PREVENT (2024) suggested preventive PCI of such plaques reduces events, but routine treatment is not guideline-endorsed.

Q130. B. Glagov compensatory (positive) remodeling: the vessel enlarges to preserve the lumen until plaque burden reaches about 40%, so angiography underestimates atherosclerosis

Glagov showed that the external elastic membrane enlarges in proportion to plaque until plaque occupies about 40% of the vessel area, after which the lumen narrows. Positive remodeling characterizes ACS and vulnerable lesions; negative (constrictive) remodeling is seen in stable, diabetic and restenotic lesions. Angiography is a lumenogram and therefore underestimates atheroma burden, and the 'normal' reference segment is usually diseased.

Complications

Q22. C. Reinflate a balloon at the perforation site at low pressure to occlude it

For an Ellis type III perforation, immediate proximal/at-site balloon occlusion stops extravasation while you assess hemodynamics, obtain echo, prepare a covered stent (via a second guide if needed), and arrange pericardiocentesis if tamponade develops. Protamine comes after definitive sealing.

Q23. C. Distal wire perforation of a small branch

Delayed tamponade after PCI classically results from an unrecognized distal wire perforation, particularly with hydrophilic/polymer-jacketed wires in tortuous vessels. Treat with pericardiocentesis, reversal, and angiography to seal the branch (coils/fat/microspheres) if bleeding continues.

Q24. C. Type C; stent

NHLBI type C is an extraluminal contrast cap with persistent staining; types C–F carry high risk of abrupt closure and should be stented (after confirming the wire is in the true lumen, ideally with IVUS/OCT). Types A–B without flow limitation can be observed.

Q25. B. Stent under-expansion

Early ST is mechanical: under-expansion, edge dissection, residual disease/geographic miss, plus DAPT failure. Neoatherosclerosis, late malapposition and hypersensitivity drive late/very late ST. Image every ST and post-dilate rather than adding layers.

Q26. C. Non-contrast CT abdomen/pelvis, fluid resuscitation and type-and-cross, with low threshold for contralateral covered stent

A high (above the inguinal ligament/inferior epigastric) puncture with hypotension, tachycardia and flank/back pain is retroperitoneal hemorrhage until proven otherwise. CT confirms; management is resuscitation, reversal, and endovascular balloon tamponade/covered stent from the contralateral side if unstable; surgery is rarely needed.

Q27. B. Heparin ≥ 5000 U and patent hemostasis reduce its incidence

RAO (2–10%) is mostly asymptomatic but sacrifices the access site and graft conduit. Prevention: adequate heparin, patent hemostasis, shortest compression, smallest sheath, distal radial access. Early ulnar compression can recanalize. Allen/Barbeau testing did not predict complications in RIVAL.

Q28. B. Adenosine release from microparticle embolization; prophylactic temporary pacing wire

Rotablation of the RCA or a dominant LCx frequently causes bradycardia/heart block attributed to adenosine released from microparticles and slow-flow. A prophylactic temporary pacer (or aminophylline/atropine readiness) is recommended for these vessels.

Q29. B. Cholesterol embolization syndrome

Cholesterol (atheroembolic) syndrome follows aortic instrumentation: delayed AKI, livedo, blue toes with intact pulses, eosinophilia, low complement. Treatment is supportive; avoid further instrumentation; statins; do not intensify anticoagulation.

Q126. B. Advance a 1.5 mm balloon through the stent on the retained wire, inflate it distal to the stent, and withdraw stent and balloon together into the guide

Stent loss (~0.2–0.3%) occurs with calcified tortuous vessels, direct stenting, or pulling an undeployed stent back into the guide. With the wire still through the stent: pass a small balloon through it, inflate distally and withdraw the assembly into the guide (two-wire twisting is an alternative). If the wire is lost: snare, and if retrieval fails deploy it in place or crush it against the wall with another stent (acceptable in small branches). Never leave a stent free in the aorta or left main; stents embolized to the periphery are usually benign.

Q127. B. 100% oxygen, aspiration through the guide or an aspiration catheter, hemodynamic support, and forceful saline injection or balloon dilation to disperse the air if needed

Coronary air embolism presents as abrupt chest pain, ST elevation, hypotension, heart block or VF, with radiolucent filling defects. Treat with 100% oxygen (speeds nitrogen absorption), aspiration, pressors, pacing or CPR as needed, and intracoronary flushing or balloon dilatation to break up large bubbles. Prevention: aspirate and flush at every catheter exchange, keep the manifold bubble-free, and never inject without a visible pressure tracing.

Q128. C. Keep the guide position, avoid forceful injections, wire the true lumen with a workhorse wire (IVUS if unsure), and stent the ostial LM entry point; if the dissection extends into the aorta, obtain CT and involve surgery

Catheter-induced LM or RCA dissection is more frequent with Amplatz shapes, deep engagement, non-coaxial alignment and ostial disease. Keep the guide position, avoid injections, wire the true lumen, and stent the entry point. Retrograde aortic extension (Dunning class I <40 mm, II >40 mm, III beyond the ascending aorta) is managed by sealing the entry point for classes I–II and surgery for class III. IVUS confirms the wire is in the true lumen.

Hemodynamics and mechanical support

Q30. C. 2.0

Mixed venous (Flamm) = $(3 \times 68 + 74)/4 = 69.5\%$. $Q_p/Q_s = (SaO_2 - MVO_2)/(PvO_2 - PaO_2) = (97 - 69.5)/(98 - 83) = 27.5/15 \approx 1.8-2.0$ (assuming PvO_2 98%). The step-up at the RA ($\geq 7\%$) indicates an atrial-level shunt; $Q_p/Q_s \geq 1.5$ with RV enlargement is an indication for closure.

Q31. B. Constrictive pericarditis

Equalized diastolic pressures (within 5 mmHg), RVSP < 55 , dip-and-plateau, and above all respiratory DISCORDANCE of LV/RV systolic pressures (ventricular interdependence; systolic area index > 1.1) define constriction. Restriction shows concordance, LVEDP–RVEDP > 5 and RVSP > 55 .

Q32. C. Blunted/absent y descent with equalization of RA, RV diastolic, PA diastolic and PCWP

In tamponade, the pericardial pressure prevents early diastolic filling, so the y descent is lost, while diastolic pressures equalize and pulsus paradoxus develops. Prominent y descent and Kussmaul suggest constriction.

Q33. A. CPO 0.4 W, PAPi 2.0: LV-dominant shock with poor prognosis

$CPO = MAP \times CO / 451 = 60 \times 3 / 451 \approx 0.40$ W (< 0.6 W = strongest mortality predictor in SHOCK). $PAPi = (30 - 14)/8 = 2.0$ (> 0.9 argues against RV failure). High PCWP with low CPO is LV-dominant shock; consider LV unloading support.

Q34. B. Brockenbrough–Braunwald–Morrow sign; dynamic obstruction as in HOCM

In HOCM the post-extrasystolic potentiation increases contractility and obstruction, so the gradient rises and aortic pulse pressure falls. In fixed AS, pulse pressure increases after a PVC.

Q35. B. Moderate-to-severe aortic regurgitation

IABP is contraindicated in significant aortic regurgitation (diastolic inflation worsens regurgitation), aortic dissection, and severe aortoiliac disease (relative). AF makes timing difficult but is not a contraindication.

Q36. B. Add an LV venting strategy such as Impella or IABP

Retrograde ECMO flow increases LV afterload; a non-ejecting LV distends, causes pulmonary edema and LV thrombus. Venting (Impella 'ECPELLA', IABP, septostomy, or LA cannula) is indicated. Increasing flow worsens afterload.

Q37. B. Differential hypoxemia (Harlequin syndrome): the recovering native heart ejects poorly oxygenated blood to the upper body

With lung failure and recovering LV function, native output supplies the aortic arch (brain, coronaries, right arm) with deoxygenated blood while the ECMO blood reaches the lower body. Monitor the right radial; fix by improving lung ventilation/oxygenation or converting to VAV/central ECMO.

Q91. C. Deliberate temperature control between 32 and 37.5 °C for at least 36 hours with fever avoidance

TTM2 showed hypothermia at 33 °C was no better than normothermia with fever prevention. The 2025 AHA guideline recommends deliberate temperature control (32–37.5 °C) for ≥ 36 h in comatose post-arrest patients, avoiding fever.

Q92. C. Closure may be considered at an expert center (PVR 2 to <5 WU), possibly after vasodilator therapy or with a fenestrated device

The 2025 ACHD guideline: $PVR \leq 2$ WU → close; PVR 2 to <5 WU → consider closure in expert centers (treat-and-repair, test occlusion, fenestrated device); high PVR /Eisenmenger → no closure. Also test balloon occlusion in older patients with diastolic dysfunction because the ASD may be acting as a pop-off.

Q136. B. True severe low-flow low-gradient AS with contractile reserve; aortic valve replacement (TAVR or SAVR) is indicated

Classical low-flow low-gradient AS ($EF < 50\%$, $AVA \leq 1.0$ cm², mean gradient <40 mmHg): dobutamine stress distinguishes true severe (AVA stays ≤ 1.0 cm² while the gradient rises to ≥ 40 mmHg) from pseudo-severe (AVA rises above 1.0–1.2 cm²). Contractile reserve (stroke volume increase $\geq 20\%$) predicts better surgical outcome, but its absence is not a contraindication (TOPAS); CT calcium scoring (men >2000, women >1200 AU) is an alternative. Paradoxical low-flow (preserved EF , $SVI < 35$) requires excluding measurement error.

Structural heart

Q38. B. Coronary obstruction; coronary protection with a wire/undeployed stent or BASILICA

Low coronary height (<10–12 mm), narrow sinuses, and long/bulky leaflets predict coronary obstruction (mortality ~40%). Protect with a wire and undeployed stent in the coronary ('chimney/snorkel'), consider leaflet laceration (BASILICA) or surgery.

Q39. B. Pre-existing right bundle branch block

Pre-existing RBBB is the strongest predictor of high-grade AV block after TAVR (the left bundle is injured by the prosthesis in the LVOT). Other predictors: self-expanding valve, deep implantation, short membranous septum, LVOT calcium.

Q40. B. Aspirin alone

POPular TAVI cohort A: aspirin alone reduced bleeding without increasing ischemic events versus DAPT. GALILEO showed harm with rivaroxaban. If OAC is indicated for another reason, OAC alone (cohort B).

Q41. B. Percutaneous mitral balloon valvuloplasty

Severe symptomatic MS with favorable anatomy (Wilkins ≤ 8 , no commissural calcification), <2+ MR, and no LA thrombus is a class 1 indication for PMBV. Moderate-severe MR or LA thrombus are contraindications.

Q42. B. EF 30%, LVESD 55 mm, EROA 40 mm², symptomatic on maximally tolerated GDMT

COAPT enrolled EF 20–50%, $LVESD \leq 70$ mm, $EROA \geq 30$ mm² with symptoms despite GDMT optimized by a HF specialist ('disproportionate' MR) and showed reduced HF hospitalization and mortality. MITRA-FR-like patients (larger LV, smaller EROA, less GDMT) did not benefit. A mean gradient >5 mmHg is a relative contraindication.

Q43. B. Posterior and superior, 4–5 cm above the annulus

TEER requires a high (4–5 cm; 3.5–4 cm for medial jets) posterior-superior puncture for adequate height to maneuver the clip above the leaflets. LAAO uses an inferior-posterior puncture.

Q44. C. Percutaneous PFO closure plus antiplatelet therapy

RESPECT (long-term), CLOSE, REDUCE and DEFENSE-PFO show ~50% relative reduction in recurrent stroke with closure plus antiplatelet therapy in patients ≤ 60 with high-risk PFO features. Peri-procedural AF (~5%) is the main adverse effect.

Q45. C. Do not close now; PVR/SVR ratio >2/3 and PAP >2/3 systemic; consider vasodilator testing/PH therapy and reassess

Closure is contraindicated when PVR >2/3 SVR or PA pressure >2/3 systemic without demonstrated reversibility (vasodilator or balloon occlusion testing) because the defect is acting as a 'pop-off' for a failing RV (Eisenmenger physiology). The 2025 ACHD guideline frames it by PVR: ≤ 2 WU close, 2 to <5 WU consider in expert centers, higher no closure. Treat PH and reassess ('treat and repair').

Q46. B. Device erosion of the atrial roof/aorta

Device erosion (0.1–0.3%) is associated with deficient aortic rims and oversizing; it presents from days to years with pericardial effusion/tamponade or aorto-atrial fistula, requiring surgical removal.

Q47. B. Warfarin/DOAC plus aspirin for 45 days, TEE, then DAPT to 6 months, then aspirin

Watchman protocol: OAC + aspirin for 45 days, TEE to exclude device-related thrombus and peri-device leak >5 mm, then DAPT until 6 months, then aspirin indefinitely. DAPT-only regimens are used in patients who cannot take OAC.

Q48. B. RBBB; about 10%

ASA infarcts the basal septum supplied by the first septal perforator, typically producing RBBB (myectomy produces LBBB). Complete heart block needing PPM occurs in roughly 10%, higher with pre-existing LBBB.

Q85. B. TF-TAVR reduced death/stroke/unplanned CV hospitalization vs surveillance, and TAVR is FDA-approved for asymptomatic severe AS

EARLY TAVR (2024) randomized asymptomatic severe AS to TF-TAVR vs clinical surveillance and showed fewer deaths/strokes/unplanned CV hospitalizations with early TAVR; the FDA approved TAVR for asymptomatic severe AS in May 2025. AVATAR/EVOLVED support early SAVR. ESC/EACTS 2025 now favors TAVI at ≥ 70 years.

Q86. C. JenaValve Trilogy

JenaValve Trilogy (ALIGN-AR) was approved in March 2026 as the first TAVR for aortic regurgitation; it anchors on the leaflets rather than annular calcium. Off-label TAVR in pure AR otherwise risks embolization.

Q87. B. PCI of significant lesions (FFR ≤ 0.80 or $\geq 90\%$ stenosis, proximal, ≥ 2.5 mm) reduced death/MI/urgent revascularization at 2 years

ACTIVATION (2021) failed non-inferiority with more bleeding for routine pre-TAVR PCI, but NOTION-3 (2024) showed PCI of physiologically or angiographically significant proximal lesions reduced death/MI/urgent revascularization (with more bleeding). Coronary evaluation before TAVR is Class 1; complex multivessel disease (SYNTAX >32) favors SAVR + CABG.

Q88. C. CHAMPION-AF: non-inferior for CV death/stroke/SE with less non-procedural bleeding but numerically more ischemic stroke; CLOSURE-AF: LAAO failed non-inferiority

CHAMPION-AF (Watchman FLX vs DOAC in OAC-eligible AF): non-inferior for CV death/stroke/SE (5.7% vs 4.8%), less non-procedural bleeding (10.9% vs 19.0%), ischemic stroke/SE numerically higher (3.2% vs 2.2%). CLOSURE-AF (HBR/OAC-contraindicated vs best medical care) failed non-inferiority. LAAO remains 2a with a contraindication to long-term OAC. NCDR data show any peri-device leak, even <5 mm, associates with more ischemic events.

Peripheral and noncardiac vascular

Q65. B. Carotid endarterectomy

CREST: periprocedural stroke was higher with CAS and MI higher with CEA; outcomes favored CEA in patients >70 years. CAS is preferred for high surgical-risk anatomy (prior neck surgery/radiation, contralateral occlusion, high lesion, restenosis). TCAR is an alternative with lower stroke risk than transfemoral CAS.

Q66. B. Stenting should be reserved for flash pulmonary edema, refractory hypertension with bilateral stenosis, or progressive renal failure

ASTRAL and CORAL showed no benefit of routine stenting over medical therapy. Selected indications remain. FMD is treated with balloon angioplasty alone; ostial atherosclerotic lesions need stents when intervened.

Q67. B. It reduces hemodynamic decompensation but increases major and intracranial bleeding without a mortality benefit; reserve for deterioration

PEITHO: tenecteplase reduced death/hemodynamic collapse at 7 days (driven by decompensation) but increased major (11.5% vs 2.4%) and intracranial bleeding (2% vs 0.2%) with no mortality benefit. Rescue lysis or catheter-directed therapy is used for deterioration.

Q68. B. Retrograde flow from lumbar or inferior mesenteric branches into the sac; treat if the sac enlarges

Type I = attachment site (treat), II = branch retrograde flow (most common; observe unless sac grows, then embolize), III = component/fabric (treat), IV = porosity (self-limited), V = endotension.

Q84. B. CAS plus intensive medical management reduced stroke/death vs medical management alone, and CMS covers CAS for asymptomatic $\geq 70\%$ stenosis regardless of surgical risk

CREST-2: in asymptomatic $\geq 70\%$ stenosis, CAS + intensive medical management reduced periprocedural stroke/death plus 4-year ipsilateral stroke (2.8% vs 6.0%, $P=0.02$); the CEA comparison was not significant (3.7% vs 5.3%). The Oct 2023 CMS NCD covers CAS for symptomatic $\geq 50\%$ and asymptomatic $\geq 70\%$ regardless of surgical risk, with shared decision-making. Patient selection and life expectancy still matter.

Q95. C. Single antiplatelet therapy for symptomatic PAD, and rivaroxaban 2.5 mg BID plus aspirin (Class 1) for symptomatic PAD and after revascularization

The 2024 PAD guideline recommends single antiplatelet therapy (aspirin or clopidogrel) for symptomatic PAD, and low-dose rivaroxaban 2.5 mg BID plus aspirin (COMPASS, VOYAGER PAD) to reduce MACE and MALE, Class 1. Full-dose OAC without another indication is harmful (WAVE); ticagrelor equaled clopidogrel (EUCLID); vorapaxar reduces acute limb ischemia but is contraindicated after stroke.

Q138. B. Surgical bypass with the single-segment great saphenous vein, which reduced major adverse limb events or death compared with endovascular therapy; without adequate vein the strategies were equivalent

BEST-CLI (2022): with an adequate single-segment GSV (cohort 1), bypass reduced MALE or death (42.6% vs 57.4%, HR 0.68), mainly through fewer major reinterventions; without adequate vein (cohort 2) outcomes were similar. BASIL-2 (infrapopliteal disease) favored an endovascular-first strategy for amputation-free survival. Choose by conduit availability, anatomy, and surgical risk.

Anatomy

Q96. A. Anomalous LCx from the right sinus with retroaortic course; benign

The circumflex from the right sinus or proximal RCA is the most common clinically relevant anomaly and almost always takes a retroaortic course (posterior dot sign), which is benign. An anterior dot in RAO suggests an interarterial course of an anomalous left coronary, which is the dangerous variant. Engage a right-sinus circumflex with JR4, AR or multipurpose catheters.

Q97. C. Surgical unroofing (Class 1 with symptoms or ischemia)

An anomalous left coronary from the right sinus with an interarterial/intramural course is the highest-risk anomaly and the second leading cause of sudden death in young athletes. Surgery (unroofing preferred) is Class 1 for symptoms or ischemia and reasonable even when asymptomatic in young/athletic patients. Bypass grafts fail from competitive flow; PCI is reserved for poor surgical candidates. Stress testing is unreliable because the obstruction is dynamic.

Q98. B. Arteria lusoria (aberrant right subclavian, ~1%); right radial access to the coronaries is difficult, so use left radial or femoral

An aberrant right subclavian artery (arteria lusoria) arises as the last arch branch and runs retro-esophageally, making catheter passage from the right arm into the ascending aorta tortuous. Switch to left radial or femoral access. A bovine arch (common origin of innominate and left CCA, 15–25%) matters for carotid stenting, not for coronary access.

Q99. B. Multipurpose (MP) or JR4

Right-sided grafts take off anteriorly and run downward, so a multipurpose or JR4 (or right coronary bypass) catheter engages coaxially. Left-sided grafts have a superior/leftward takeoff, engaged with JR4, LCB, AL1 or a hockey stick. The LIMA is engaged with an IM catheter (or JR4) from the subclavian, easiest from the left radial.

Q100. B. Persistent left superior vena cava draining to the coronary sinus

A persistent left SVC (about 0.5% of the population, more in congenital heart disease) drains into the coronary sinus, which becomes dilated. It is a common cause of an unexpected left-sided catheter course. The azygos runs posteriorly along the spine; scimitar syndrome is an anomalous right pulmonary vein to the IVC.

Q101. B. This is left dominance: the LCx gives the PDA and posterolateral branches in ~8–10% of people

Dominance is defined by which artery gives rise to the PDA and the posterolateral branches: right ~85%, left ~8%, co-dominant ~7%. In left dominance the AV nodal artery also usually arises from the circumflex, so LCx occlusion can cause heart block. Collateral filling of the PDA via the LAD apex does not define dominance.

Q102. B. ALCAPA (Bland-White-Garland), adult form; surgical reimplantation of the left coronary

ALCAPA presents in infancy (85–90%) with heart failure, but adult survivors have extensive RCA-to-LCA collaterals with steal into the pulmonary artery, causing ischemia, MR, VT and sudden death. Treatment is surgical reimplantation to establish a two-coronary system. A coronary fistula shows a dilated feeding vessel draining directly into a chamber (>90% right-sided).

Q116. B. Antithrombotic therapy (antiplatelet ± anticoagulation) and thrombus management; avoid stenting ectatic segments; screen for abdominal aortic aneurysm

Coronary ectasia ($\geq 1.5 \times$ reference) causes slow flow and thrombus formation; ACS in ectatic vessels is managed medically with antithrombotics and thrombus aspiration/GPI as needed. Stents in ectatic segments malappose and thrombose. Covered stents or surgery are reserved for discrete large or enlarging aneurysms. Ectasia is associated with abdominal aortic aneurysm; Kawasaki giant aneurysms (≥ 8 mm) receive anticoagulation plus aspirin.

Radiation safety

Q69. B. Document the dose, counsel the patient about possible skin effects, and arrange skin examination in 2–4 weeks

A substantial radiation dose level (air kerma ≥ 5 Gy, KAP ≥ 500 Gy·cm², peak skin dose ≥ 3 Gy or fluoro time ≥ 60 min) requires documentation, counseling and follow-up of the skin (erythema at 2 Gy appears within days-weeks; delayed effects later). Air kerma > 15 Gy (peak skin dose) is the Joint Commission sentinel event threshold.

Q70. A. Switching from 15 to 7.5 frames per second

Reducing the pulse/frame rate roughly halves dose. Steep LAO views put the tube on the operator's side and increase scatter; lowering the table increases skin entrance dose; magnification and open collimation increase dose.

Q71. B. Fetal dose limit is 5 mSv for the pregnancy and 0.5 mSv per month, measured with a badge under the apron at the waist

NCRP: 5 mSv (0.5 rem) total for the declared pregnancy, 0.5 mSv/month, monitored by a waist badge under the apron. With standard shielding, fetal doses are typically far below this and continued work is permitted.

Q72. C. Reference-point air kerma

Reference-point air kerma (Gy at the interventional reference point) is the surrogate for peak skin dose and deterministic injury. KAP relates to stochastic risk (total energy imparted); fluoroscopy time is the poorest metric.

Statistics, quality and regulation

Q73. D. 25

ARR = 12% – 8% = 4%; NNT = $1/0.04 = 25$. RRR = $4/12 = 33\%$.

Q74. B. Non-inferiority NOT established because the upper CI bound exceeds the margin

Non-inferiority requires the entire CI (its upper bound) to lie below the margin. $2.8\% > 2.5\%$, so non-inferiority is not shown. Both ITT and per-protocol analyses should support non-inferiority since ITT biases toward the null.

Q75. D. 33%

Per 1000 patients: 100 diseased → 90 TP; 900 healthy → 180 FP. PPV = $90/(90+180) = 33\%$. PPV falls with lower prevalence; NPV here = $720/(720+10) \approx 99\%$.

Q76. C. Cox proportional hazards regression

Cox regression models time-to-event data with covariates and yields hazard ratios. Kaplan–Meier/log-rank compares unadjusted survival curves; logistic regression treats the outcome as binary without time.

Q77. B. Performance and detection (ascertainment) bias, including placebo effect

Randomization handles selection bias; blinding/sham controls handle performance/detection bias and placebo response, which matter for subjective endpoints like angina and exercise time.

Q78. B. Probability of detecting a true difference of the pre-specified size ($1 - \beta$)

Power = $1 - \beta$ = probability of rejecting the null when a true effect of the assumed size exists. Alpha is the type I error rate. Underpowered trials produce false negatives and imprecise estimates.

Q109. B. Outlier status often reflects documentation or abstraction errors, so audit the data before changing practice; NCDR metrics include risk-adjusted mortality, bleeding, AKI, 30-day readmission, discharge medications, radiation dose and AUC

CathPCI metrics: risk-adjusted mortality, bleeding, AKI, readmission, discharge aspirin/P2Y12/statin, radiation dose, cardiac rehab referral, door-to-device, and AUC rating. Apparent outliers frequently stem from data abstraction problems. For AUC, 'rarely appropriate' includes asymptomatic 1–2 vessel disease without proximal LAD, no antianginals and low-risk findings; 'severe' means $\geq 70\%$ non-LM ($\geq 50\%$ LM) or 40–69% with FFR ≤ 0.80 / iFR ≤ 0.89 .

Q112. B. The ICER is \$80,000 per QALY, in the intermediate-value range by common US thresholds (\$50,000–\$150,000 per QALY)

ICER = incremental cost / incremental QALY = $40,000 / 0.5 = \$80,000$ per QALY. ACC/AHA value statements: high value $< \$50,000$, intermediate $\$50,000$ – $\$150,000$, low value $> \$150,000$ per QALY gained. A dominant strategy is both cheaper and more effective; cost-effective does not mean cost-saving. QALY = life-years $\times$ utility (0–1).

Q113. B. Report the device-related death to both the FDA and the manufacturer within 10 working days under the Safe Medical Devices Act

Under the Safe Medical Devices Act, user facilities must report device-related deaths to the FDA and the manufacturer within 10 working days, and serious injuries to the manufacturer (or FDA if unknown). Clinician MedWatch reporting is voluntary. Reports populate the MAUDE database, a passive surveillance system without denominators. Stents and valves are Class III devices approved through PMA with IDE trials; most catheters are Class II via 510(k).

Q137. B. Per-protocol (or as-treated) analysis should be reported alongside ITT because crossover and non-adherence dilute differences in ITT and bias toward non-inferiority

In non-inferiority designs, ITT dilutes true differences and biases toward 'no difference', so non-inferiority should be shown in both ITT and per-protocol populations. The pre-specified margin (δ) must be justified from the active control's known effect; non-inferiority is established when the confidence-interval bound for the difference excludes δ . Trials may test non-inferiority then superiority hierarchically, but a non-significant superiority test never establishes non-inferiority.